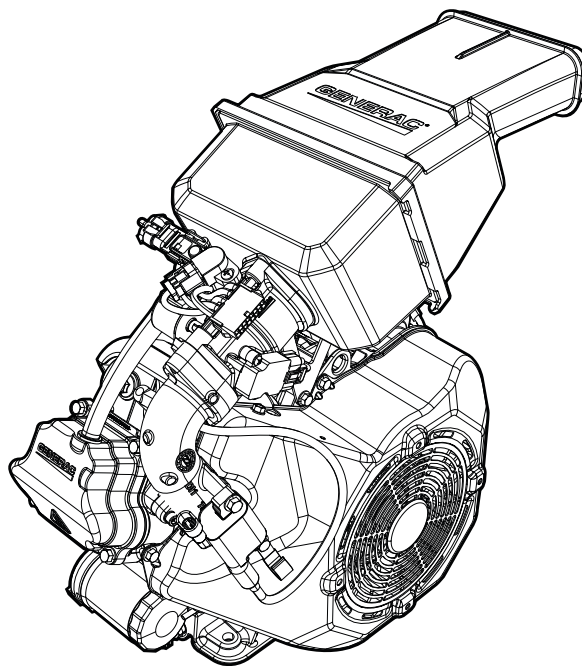


Service Manual

459cc Engine



SAVE THIS MANUAL FOR FUTURE REFERENCE

 CALIFORNIA WARNING

This product can expose you to chemicals including benzene, a carcinogen and reproductive toxicant, which are known to the State of California to cause cancer and birth defects or other reproductive harm.

For more information, go to:
www.p65warnings.ca.gov

(P000808)

Table of Contents

Section 1: Forward

Introduction	5
Intended Use	5
Service Information	5
Scope	5

Section 2: Safety Rules & General Information

Notice to Users	7
Read This Manual Thoroughly	7
Emissions	7
Safety Rules	7
Non-Generac Parts and Equipment	7
General Hazards	8
Replacement Parts	8
Exhaust Hazards	9
Electrical Hazards	9
Fire and Fuel Hazards	9
Hearing Loss Hazard	10
Moving Parts Hazard	10
Before Starting Equipment	10
Operating Equipment	11
Transporting or Repairing Equipment	11
Hot Surfaces Hazard	11
Spark Hazards	11
When Testing For Engine Spark	11
Battery Hazards	11
General Maintenance	12

Section 3: Four-Cycle Engine Theory

Introduction	13
(A) Intake	13
(B) Compression	13
(C) Power	13
(D) Exhaust	13

Section 4: General Information

Maintenance Recommendations	15
Checking Engine Oil	15

Engine Oil	15
Inspecting Engine Oil Level	15
Changing Oil and Oil Filter	15
Inspecting and Cleaning Intake Screen	16
Replacing Spark Plugs	16
Maintaining Air Cleaner	17
Troubleshooting	17
Systematic Inspection	17
Systematic Inspection	17
Checking Fuel Pressure	17
Checking Ignition	17
Checking Oil Pressure	17
Cylinder Leakdown Test	18
Cylinder Leakdown Test	19
Testing Procedure	19
Results	19
Factors Affecting Cylinder Condition	19

Section 5: Specifications

Specifications	21
Torque Specifications	21
Torque Sequences	23
Component Weights	23
Approved Sealants/Lubricants	24

Section 6: Engine Disassembly

Disassembling Long Block	25
Long Block Disassembly	25
Removing Ignition Coil and Spark Plug	25
Removing Spark Plug and Starter	25
Removing Air Box	25
Removing Breather Hose From Rocker Cover	26
Removing Mixer Assembly	26
Removing Intake and Exhaust Components	26
Removing Oil Drain Line	26
Removing Oil Filter and Low Oil Pressure Switch	27
Removing Cylinder Wrappers and Blower Housing	27
Removing Flywheel	27
Removing Rocker Covers	28
Removing Pushrods and Rocker Arms	28
Removing Cylinder Head	28
Disassembling Cylinder Head	29

Section 7: Ignition

Inspecting Ignition	31
Removing Ignition Coil	31
Testing Ignition Coil	31
Inspecting Wire Harness	32
Flywheel	32
Removing Flywheel and Fan.....	32
Installing Flywheel and Fan	32
Installing Crankshaft Position Sensor	33
Checking Spark Plug Gap	33
Installing Ignition Coil	33

Section 8: Sub-Assemblies

Gear Cover Sub-Assembly	35
Cylinder Head Sub-Assembly	35

Section 9: Cylinder Head and Valves

Inspecting and Repairing Cylinder Head	37
Measuring Valves and Springs	37
Gasket Removal	38
Assemble Cylinder Head.....	38

Section 10: Electric Starter

General Information.....	39
Troubleshooting.....	39
Starter Troubleshooting.....	39
Engine cranks slowly:.....	39
Engine will not crank if:.....	39
Starter motor spins but does not crank engine:.....	39
Starter motor spins but will not stop:	39
Test Equipment.....	39
Digital Multimeter (DMM)	39
Testing the Starter Motor	40
Checking the Pinion	40
Tools for Starter Performance Test	40
Testing Starter Performance	40
Verifying Starter Voltage Drop.....	40
Testing Starter Amp Draw	41
Battery Information	41
Battery Cable Recommendations	41
Battery Cable Recommendations:	41
Inspecting Battery.....	42

Battery Disposal	42
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Section 11: Lubrication System

Description	43
Protection Systems	43
Low Oil Pressure Switch.....	43
Checking Engine Oil Level	43
Changing Engine Oil And Filter	44

Section 12: Engine Assembly

Long Block Assembly	45
Installing Gear Cover	45
Installing Cylinder Head.....	45
Installing Valve Train	46
Installing Rocker Cover	47
Installing Ground Wire and Stator.....	47
Installing Cylinder Wrappers	48
Installing Blower Housing	48
Installing Oil Filter and Oil Pressure Switch	48
Installing Oil Drain Line and Oil Dipstick.....	48
Installing Spark Plugs.....	49
Installing Starter.....	49
Intake and Exhaust Assembly	49
Installing Intake Manifold.....	49
Installing Head Temperature Sensor	49
Installing Throttle Body and Air Box.....	50
Installing Air Filter	50
Installing Exhaust Manifold.....	50

Section 13: Troubleshooting

Section 1: Forward

Introduction

This manual has been written and published by Generac Power Systems, Inc. ("Generac®") to aid dealers, mechanics, and company service personnel when servicing products described herein.

It is assumed these personnel are familiar with servicing procedures for these products, or similar products, manufactured and marketed by Generac. It is also assumed they have been trained in the recommended servicing procedures for these products, which includes the use of mechanics' hand tools and any special tools required.

Correct service and repair is important to the safe, economical, and reliable operation of the products described herein. Troubleshooting, testing, service, and repair procedures recommended by Generac and described in this manual are effective methods of performing such operations. Some of these operations or procedures may require use of specialized equipment. Such equipment should be used when and as recommended.

Generac could not possibly know of and advise the service trade of all conceivable procedures or methods by which a service might be performed, nor of any possible hazards and/or results of each procedure or method. Generac has not undertaken any such wide evaluation. Therefore, anyone who uses a procedure or method not recommended by the manufacturer must first satisfy themselves of neither their safety, nor the safety of those around them or the product's safety will be endangered by the service or operating procedure selected.

Some components or assemblies of the product described in this manual may not be considered repairable. Disassembly, repair, and reassembly of such components may not be included in this manual.

Engines described herein may be used to power a wide variety of products. Service and repair instructions relating to any such products are not covered in this manual. For information pertaining to use of these engines with other products, see owner's or service manuals pertaining to said products.

The information in this manual is accurate based on products produced at the time of publication. The manufacturer reserves the right to make technical updates, corrections, and product revisions at any time without notice.

Intended Use

This engine was designed and manufactured for specific applications. Do not attempt to modify the equipment or use it for any application for which it was not designed.

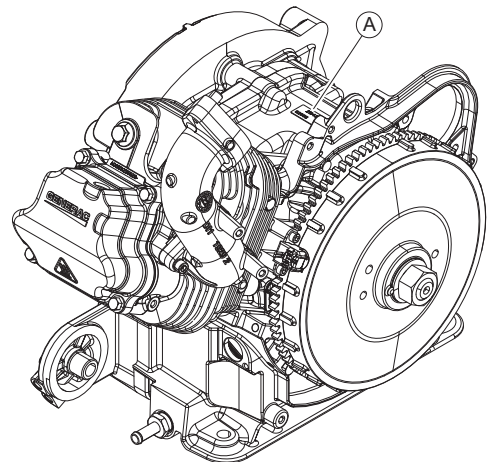
Generac does not approve or authorize use of these engines on all-terrain vehicles (ATVs), go-carts, motorbikes, aircraft products, personal watercraft, or vehicles intended for use in competitive events. Use of this product in any other than its intended application will void the warranty. Use of these engines in such applications could result in property damage, serious injury (including paralysis), or even death. Do not remove or otherwise alter any safety devices on this engine or any unit it is installed into.

Service Information

Service on this engine within and after the warranty period can be performed by any Independent Authorized Service Dealer (IASD). IASDs are factory trained and capable of handling all service needs.

Always supply the complete model number and serial number of a unit as given on its data plate decal (A) when contacting an IASD about parts and service.

A warranty statement for this engine is included with the documents provided with the unit.



000048822

Figure 1-1. Data Plate Decal

Scope

Servicing information provided in this manual applies only to the Generac engine itself. Topics such as air cleaner maintenance, fuel tank maintenance, control panel, and unit rebuild are not included. For servicing information on those components, see diagnostic manual for the associated Generac product.

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Section 2: Safety Rules & General Information

Notice to Users

Study these safety rules carefully before operating or servicing this equipment. Become familiar with the owner's manual and with the engine. The engine can operate safely, efficiently, and reliably only if it is correctly operated and maintained. Many accidents are caused by failing to follow simple and fundamental rules or precautions.

The information in this manual is accurate based on products produced at the time of publication. The manufacturer reserves the right to make technical updates, corrections, and product revisions at any time without notice.

Read This Manual Thoroughly



⚠ WARNING

Consult Manual. Read and understand manual completely before using product. Failure to completely understand manual and product could result in death or serious injury. (W000100)

If any portion of this manual is not understood, contact the nearest Independent Authorized Service Dealer (IASD) for starting, operating, and servicing procedures. The owner is responsible for correct maintenance and safe use of the unit.

This manual must be used in conjunction with all other supporting product documentation supplied with the product.

IMPORTANT SAFETY INSTRUCTIONS. SAVE THESE INSTRUCTIONS for future reference. This manual contains important instructions which must be followed during placement, operation, and maintenance of the unit and its components. Always supply this manual to any individual who will use this unit, and instruct them on how to correctly start, operate, and stop the unit in case of emergency.

Emissions

The United States Environmental Protection Agency (US EPA) (and California Air Resources Board (CARB), for engines/equipment certified to California standards) requires this engine/equipment to comply with exhaust and evaporative emissions standards. Locate the emissions compliance decal on the engine to determine applicable standards. See the included emissions warranty for emissions warranty information. Follow the maintenance specifications in this manual to verify the

engine complies with applicable emissions standards for the duration of the product's life.

Safety Rules

The manufacturer cannot anticipate every possible circumstance that might involve a hazard. The alerts in this manual, and on tags and decals affixed to the unit, are not all inclusive. If using a procedure, work method, or operating technique that the manufacturer does not specifically recommend, verify that it is safe for others and does not render the equipment unsafe.

Throughout this publication, and on tags and decals affixed to the unit, DANGER, WARNING, CAUTION, and NOTE blocks are used to alert personnel to special instructions about a particular operation that may be hazardous if performed incorrectly or carelessly. Observe them carefully. Alert definitions are as follows:

⚠ DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious injury. (D000001)

⚠ WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious injury. (W000002)

⚠ CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury. (C000003)

NOTE: Notes contain additional information important to a procedure and will be found within the regular text of this manual.

These safety alerts cannot eliminate the hazards that they indicate. Common sense and strict compliance with the special instructions while performing the action or service are essential to preventing accidents.

Non-Generac Parts and Equipment

Generac products obtain listings and certifications through an evaluation of the original design and configuration of the product. Any modification to this product, including the incorporation of non-Generac parts or equipment, may void this product's listing or certification. Damage or failures caused by any

modification to the product are not covered by the product's warranty. See limited warranty for more information.

General Hazards



Automatic start-up. Disconnect utility power and render unit inoperable before working on unit. Failure to do so will result in death or serious injury.
(D000191)



Asphyxiation. Running engines produce carbon monoxide, a colorless, odorless, poisonous gas. Carbon monoxide, if not avoided, will result in death or serious injury.
(D000103)



Electrocution. High voltage is present at transfer switch and terminals. Contact with live terminals will result in death or serious injury.
(D000129)



Accidental Start-up. Disconnect the negative battery cable, then the positive battery cable when working on unit. Failure to do so could result in death or serious injury.
(W000130)



Equipment damage. Only qualified service personnel may install, operate, and maintain this equipment. Failure to follow proper installation requirements could result in death, serious injury, and equipment or property damage.
(W000182)



Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury.
(W000111)



Moving Parts. Do not wear jewelry when starting or operating this product. Wearing jewelry while starting or operating this product could result in death or serious injury.
(W000115)



Equipment and property damage. Do not alter construction of, installation, or block ventilation for generator. Failure to do so could result in unsafe operation or damage to the generator.
(W000146)



Risk of injury. Do not operate or service this machine if not fully alert. Fatigue can impair the ability to service this equipment and could result in death or serious injury.
(W000215)



Accidental Start-Up. Disconnect spark plug wires when working on unit. Failure to do so could result in death or serious injury.
(W000141)

Replacement Parts

This equipment is designed to be used with Generac authorized parts ONLY. User assumes all risks and liabilities if equipment is used with parts which DO NOT comply with minimum specifications. During any maintenance procedure, replacement fasteners must have the same measurements and strength as the removed fasteners. Metric bolts and nuts have numbers indicating their strength. Imperial bolts use radial lines to indicate strength, while most Imperial nuts do not have strength markings. Mismatched or incorrect fasteners can cause damage, malfunction, and possible injury.

Exhaust Hazards



⚠ DANGER

Asphyxiation. Running engines produce carbon monoxide, a colorless, odorless, poisonous gas. Carbon monoxide, if not avoided, will result in death or serious injury.
(D000103)



⚠ DANGER

Asphyxiation. Carbon monoxide can kill in minutes. Operate this unit outdoors only. Failure to do so will result in death or serious injury.
(D000525)



⚠ WARNING

Asphyxiation. Always use a battery operated carbon monoxide alarm indoors and installed according to the manufacturer's instructions. Failure to do so could result in death or serious injury.
(W000178)



⚠ WARNING

Asphyxiation. Do not inhale vapor or fumes. Seek medical attention immediately. Failure to do so could result in death or serious injury.
(W000342)

⚠ CAUTION

Personal injury. Wear appropriate personal protective equipment at all times while operating and servicing unit. Failure to do so could result in personal injury.
(C000419)

⚠ CAUTION

Personal injury. Wear appropriate personal protective equipment at all times while operating and servicing unit. Failure to do so could result in personal injury.
(C000419)

Electrical Hazards



⚠ DANGER

Electrocution. Water contact with a power source, if not avoided, will result in death or serious injury.
(D000104)

- Never service this engine or handle any electrical device while standing in water, while barefoot, or

while hands or feet are wet. DANGEROUS ELECTRIC SHOCK will result.

- Never operate the engine (a) in the rain; (b) in any enclosed compartment; (c) if the engine speed changes; (d) if the engine sparks; (e) if flame or smoke is observed while the engine is running.

Fire and Fuel Hazards



⚠ DANGER

Explosion and fire. Fuel and vapors are extremely flammable and explosive. Add fuel in a well ventilated area. Keep fire and spark away. Failure to do so will result in death or serious injury.
(D000105)



⚠ DANGER

Explosion and fire. Fuel and vapors are extremely flammable and explosive. No leakage of fuel is permitted. Keep fire and spark away. Failure to do so will result in death or serious injury.
(D000192)



⚠ DANGER

Explosion and Fire. Fuel and vapors are extremely flammable and explosive. Store fuel in a well ventilated area. Keep fire and spark away. Failure to do so will result in death or serious injury.
(D000143)



⚠ DANGER

Explosion and Fire. Do not fill fuel tank past full line. Allow for fuel expansion. Overfilling may cause fuel to spill onto engine causing fire or explosion, which will result in death or serious injury.
(D000214)



⚠ DANGER

Risk of fire. Allow fuel spills to completely dry before starting engine. Failure to do so will result in death or serious injury.
(D000174)



⚠ DANGER

Explosion and Fire. Drain fuel in a well ventilated area, into an approved container only after engine has cooled. Failure to do so will result in death or serious injury.
(D000180)



⚠ WARNING

Explosion and Fire. Do not smoke while refueling unit. Failure to do so could result in death, serious injury, or property or equipment damage.

(W000284)



⚠ WARNING

Fire risk. Fuel and vapors are extremely flammable. Do not operate indoors. Doing so could result in death, serious injury, or property or equipment damage.

(W000281)

⚠ CAUTION

Engine Damage. Do not operate with incorrect fuel. Doing so will result in reduced service life of the engine.
(C000569)

- Avoid spilling gasoline on a hot engine. Comply with all of the laws regulating storage and handling of gasoline.
- It is a violation of California Public Resource Code, Section 4442, to use or operate the engine on any forest-covered, brush-covered, or grass-covered land unless the exhaust system is equipped with a spark arrester, as defined in Section 4442, maintained in effective working order. Other states or federal jurisdictions may have similar laws. Contact the original equipment manufacturer, retailer, or an IASD to obtain a spark arrester designed for the exhaust system installed on this engine.

Hearing Loss Hazard



⚠ WARNING

Hearing Loss. Hearing protection is recommended when using this machine. Failure to wear hearing protection could result in permanent hearing loss.

(W000107)

Moving Parts Hazard



⚠ WARNING

Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury.

(W000111)



⚠ WARNING

Moving Parts. Do not wear jewelry when starting or operating this product. Wearing jewelry while starting or operating this product could result in death or serious injury.

(W000115)

Before Starting Equipment

⚠ WARNING

Equipment damage. Do not attempt to start or operate a unit in need of repair or scheduled maintenance. Doing so could result in serious injury, death, or equipment failure or damage.

(W000291)

⚠ WARNING

Personal Injury. Never rotate engine with spark plug removed. Rotating engine with spark plug removed could result in death or serious injury.

(W000564)

⚠ CAUTION

Engine damage. Verify proper type and quantity of engine oil prior to starting engine. Failure to do so could result in engine damage.

(C000135)

- Verify spark plug, muffler, fuel cap, and air cleaner are in place.

Operating Equipment

WARNING

Equipment damage. Operating engine with obstructed air cooling slots could result in death, serious injury, and unit damage.

(W000255)



WARNING

Fire hazard. Do not obstruct cooling and ventilating airflow around the engine. Inadequate ventilation could result in fire hazard, possible equipment damage, death or serious injury.

(W000256)

WARNING

Personal injury and equipment damage. Do not tamper with engine governed speed. Operating engine too fast or slow increases the risk of injury or permanent engine damage.

(W000254)

CAUTION

Equipment damage. Never stop the engine by moving throttle control lever to CHOKE. Doing so could result in engine damage.

(C000457)

- DO NOT tip engine or equipment at angle which causes fuel to spill.
- Reflective exhaust heat may damage the fuel tank, causing fire. Keep at least 5 ft (1.3 m) of clearance on all sides of the engine for adequate cooling, maintenance, and servicing.

Transporting or Repairing Equipment

WARNING

Accidental Start-Up. Disconnect spark plug wires when working on unit. Failure to do so could result in death or serious injury.

(W000141)

- Transport/repair with fuel tank EMPTY.

Hot Surfaces Hazard



WARNING

Hot Surfaces. When operating machine, do not touch hot surfaces. Keep machine away from combustibles during use. Hot surfaces could result in severe burns or fire.

(W000108)

- AVOID hot exhaust gases.
- Allow equipment to cool before touching.

Spark Hazards

WARNING

Accidental Start-Up. Disconnect spark plug wires when working on unit. Failure to do so could result in death or serious injury.

(W000141)

When Testing For Engine Spark



WARNING

Explosion. Turn fuel supply OFF before checking for spark. Failure to do so could result in death or severe injury.

(W000333)

WARNING

Electric Shock. Never test ignition spark with the spark plug or spark plug wire(s) removed. Always use an approved spark tester. Failure to do so could result in death, serious injury, or equipment damage.

(W000406)

- Use approved spark plug tester.
- Place the controller in MANUAL mode.

Battery Hazards



WARNING

Explosion. Do not dispose of batteries in a fire. Batteries are explosive. Electrolyte solution can cause burns and blindness. If electrolyte contacts skin or eyes, flush with water and seek immediate medical attention.

(W000162)



WARNING

Explosion. Batteries emit explosive gases while charging. Keep fire and spark away. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(W000137)



⚠ WARNING

Electrical shock. Disconnect battery ground terminal before working on battery or battery wires. Failure to do so could result in death or serious injury.

(W000164)



⚠ WARNING

Risk of burns. Batteries contain sulfuric acid and can cause severe chemical burns. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(W000138)

Always recycle batteries in accordance with local laws and regulations. Contact your local solid waste collection site or recycling facility to obtain information on local recycling processes. For more information on battery recycling, visit Call2Recycle website at: <http://Call2Recycle.org/locator>.

General Maintenance

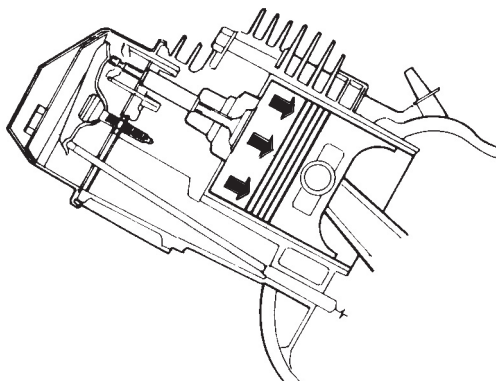
- Inspect engine periodically. Repair or replace all damaged or faulty parts immediately.
- Inspect fuel system frequently for leaks or damage. Repair or replace any damaged or leaking component immediately. Never attempt to change, alter, or modify the engine fuel system in any way which might affect safety or compliance with applicable codes and standards.

Section 3: Four-Cycle Engine Theory

Introduction

Four events must occur in the correct sequence and at the correct time for the engine to run correctly. These events are (A) intake, (B) compression, (C) power, and (D) exhaust.

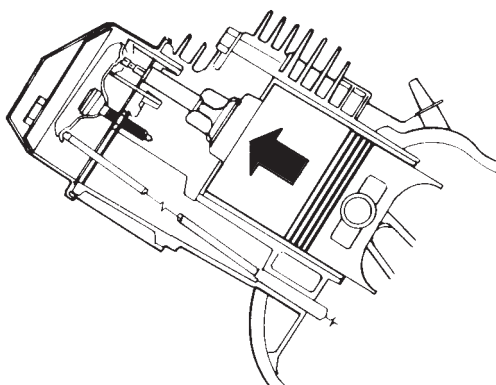
(A) Intake



001684

The piston is traveling from top dead center (TDC) to bottom dead center (BDC). The cam has opened the intake valve. The piston's downward movement in the cylinder creates a partial vacuum in the cylinder. Air at atmospheric pressure is drawn into the cylinder through the airbox and is mixed with fuel, supplied by the mixer, as it enters the cylinder. The intake stroke is complete when the piston reaches BDC.

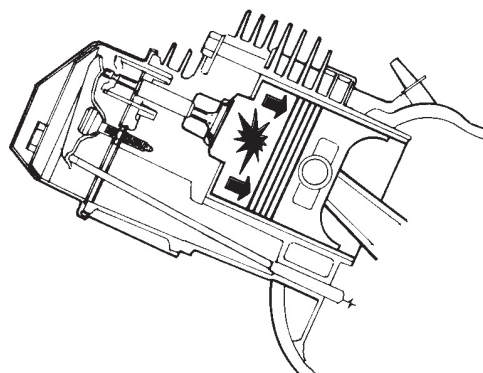
(B) Compression



001685

As the piston reaches BDC, both the intake and exhaust valves are closed. The piston moves upward toward TDC and the fuel-air mixture is compressed. Ignition occurs just before the piston reaches TDC.

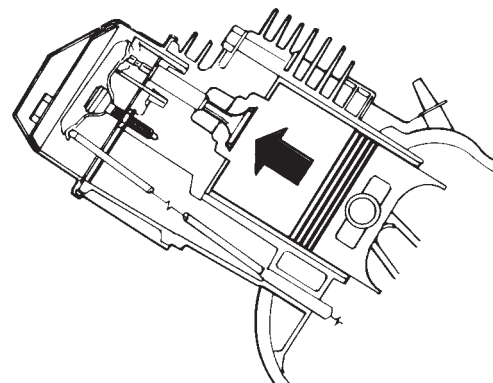
(C) Power



001686

Combustion is already in progress when the piston reaches TDC. The intake and exhaust valves remain closed as expanding gases of combustion force the piston downward.

(D) Exhaust



001687

The exhaust stroke begins when the piston has reached BDC and has started its upward movement. The intake valve is closed, and the exhaust valve is open to let gases escape.

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Section 4: General Information

Maintenance Recommendations

See the appropriate unit owner's manual for break-in and maintenance intervals, and oil and fuel recommendations.

Verify generator is in SERVICE mode via the Field Pro app or the control panel SERVICE button prior to starting any preventative or corrective maintenance.

Checking Engine Oil



Skin Irritation. Avoid prolonged or repeated contact with used motor oil. Used motor oil has been shown to cause skin cancer in laboratory animals. Thoroughly wash exposed areas with soap and water.

(W000210)

IMPORTANT NOTE: Do not overfill. Excessive oil will foam, smoke, and eventually damage internal engine parts.

See [Specifications](#) for crankcase oil capacities. Do not use special additives. Verify unit is level when filling oil.

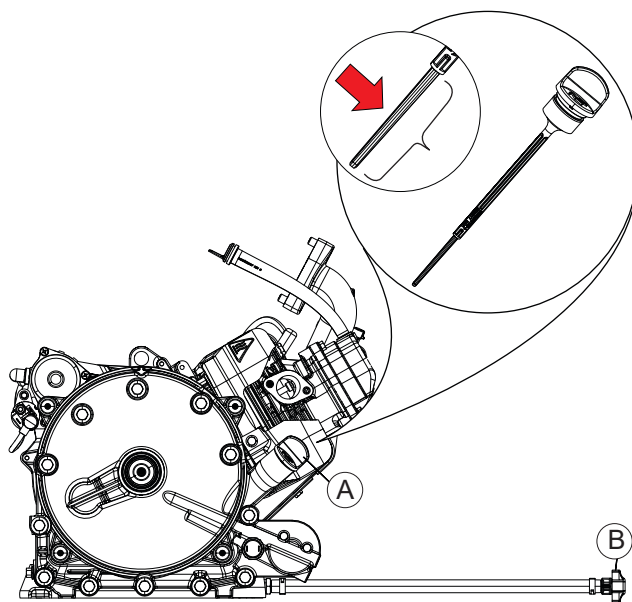
Change oil and filter after first 25 hours of operation for break-in, then resume normal maintenance interval. Change oil and filter more frequently if engine is operated in dirty or dusty conditions, operated under heavy loads, or in high ambient air temperatures.

Engine Oil

Inspecting Engine Oil Level

Proceed as follows to inspect engine oil level prior to each use, or every eight (8) hours of operation.

1. See [Figure 4-1](#). Remove oil dipstick (A) and wipe dipstick clean.



00048821

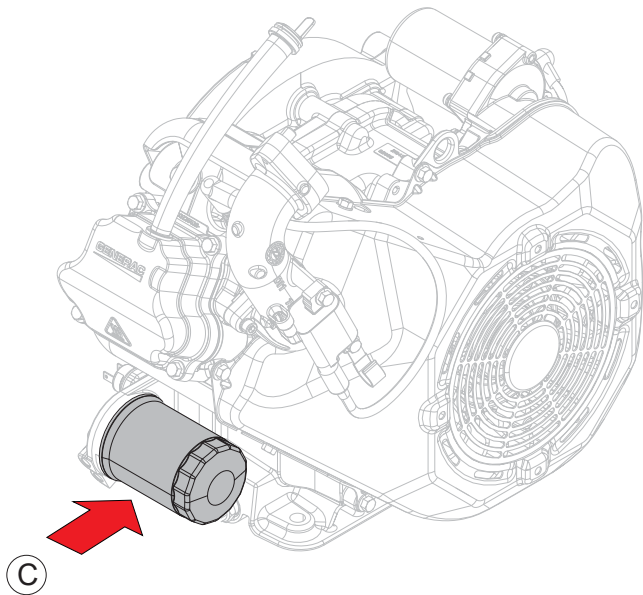
Figure 4-1. Oil Dipstick and Oil Drain Cap

2. Replace oil dipstick. Remove oil dipstick and verify oil level is within safe operating range.
3. Add recommended engine oil as necessary.
4. Install oil fill cap/dipstick and hand-tighten.

Changing Oil and Oil Filter

Proceed as follows to change oil and oil filter:

1. See [Figure 4-1](#). Remove oil drain cap (B) and drain oil while engine is still warm.
2. See [Figure 4-2](#). Remove used oil filter (C) and install oil drain cap.



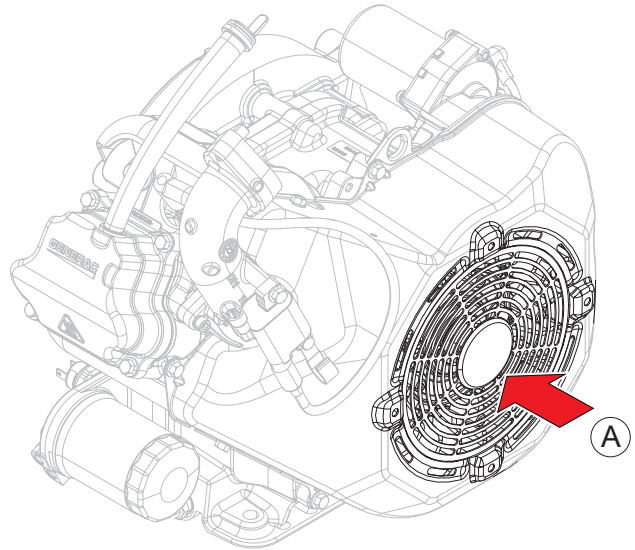
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Figure 4-2. Oil Filter

3. Lightly lubricate oil filter gasket with fresh, clean engine oil before installing new oil filter.
4. Screw filter on by hand until oil filter gasket contacts oil filter adapter. Tighten three-quarters to one full turn farther.
5. See [Figure 4-1](#). Remove oil dipstick (A) and fill slowly with new oil of required service classification and viscosity grade. Fill to bottom of threads in oil fill opening.
6. Start and run engine at idle for 30 seconds. Stop engine. Verify oil level and add if required.
7. Start engine and inspect for oil leaks.

Inspecting and Cleaning Intake Screen

See [Figure 4-3](#). Continued operation with a clogged cooling system can cause severe overheating and possible engine damage. Inspect and clean the intake screen (A) during routine and periodic maintenance.



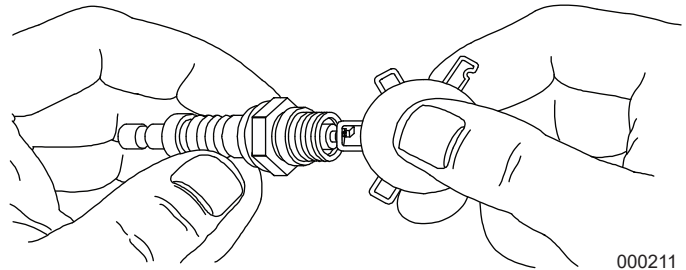
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Figure 4-3. Intake Screen

Replacing Spark Plugs

Replace spark plug if electrode is burned away or if porcelain is cracked. See generator owner's manual for spark plug replacement specification.

See [Figure 4-4](#). Inspect spark plug gap and set spark plug gap to 0.020 in (0.50 mm) for all models. Tighten spark plug to 168 in-lb (18.9 Nm).



000211

Figure 4-4. Inspecting Spark Plug Gap

Maintaining Air Cleaner



⚠ WARNING

Risk of fire. Never operate engine without the air cleaner installed. Operating engine without the air cleaner could result in death or serious injury.

(W000249)

A correctly serviced air cleaner protects internal engine parts from dirt and dust particles in the air. If air cleaner instructions are not carefully followed, dirt and dust normally collected in the air cleaner will be drawn into the engine. Particulates will wear on cylinder bores and piston rings, allowing particles to enter the crankcase and contaminate the oil, forming an abrasive mixture which will quickly wear internal moving parts.

Inspect the air cleaner on every engine serviced or repaired. If the air cleaner shows signs of neglect, show it to the user before cleaning. Instruct the user on correct care techniques to assure long engine life.

NOTE: Replace worn or damaged air cleaner gaskets and mounting gaskets to prevent dirt and dust from entering the engine due to incorrect sealing. Replace bent air cleaner mounting bracket if necessary.

Troubleshooting

Most complaints concerning engine operation can be classified as one or a combination of the following:

- Will not start
- Hard starting
- Lack of power
- Runs rough
- Vibration
- Overheating
- Noise
- High oil consumption

NOTE: What appears to be an engine malfunction may be a fault of the equipment connected to the engine rather than the engine. If equipment is suspect, see equipment affecting engine operation.

Systematic Inspection

If engine will not start and the cause of malfunction is not readily apparent, perform a systematic inspection in the following order:

1. Fuel
2. Ignition
3. Compression

This inspection, performed in a systematic manner, can usually be done in a matter of minutes. It is the quickest

and surest method of determining the cause of failure. The basic inspection procedure is the same for all engine models. Any variation, by model, will be described under the subject heading.

Systematic Inspection

Checking Fuel Pressure

- NG: 3.5 – 7.0 in H₂O (0.87 – 1.74 kPa)
- LP: 10 – 12 in H₂O (2.49 – 2.99 kPa)

For gaseous units, check the following:

- Is there sufficient fuel supply from the meter?
- Is the fuel supply pipe the correct size?
- Are there any bends or kinks in the flexible supply line?

Checking Ignition

See [Section 7: Ignition](#) for testing procedures.

Checking Oil Pressure

Proceed as follows to check oil pressure:

1. Remove generator or engine from service.
2. Inspect engine crankcase oil level. If necessary, add the recommended oil to the dipstick FULL mark. DO NOT OVERFILL.
3. See [Figure 4-5](#). Remove oil pressure switch and install an oil pressure gauge using a tee and pipe nipple between the block and original pressure gauge.

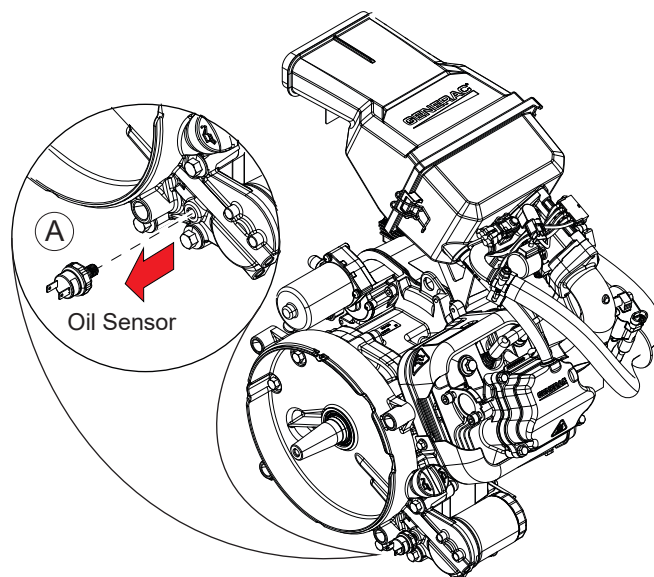


Figure 4-5. Remove Pressure Sensor

4. Thoroughly clean unit to remove any spilled oil.

00048818

- 5. Start engine while observing the oil pressure reading on the gauge. Oil pressure should be 10-50 psi (69-345 kPa) at 70 °F (21 °C).
- 6. Shut down engine.
- 7. Install original pressure sensor into sensor location on unit.
- 8. Clean area to remove any residual oil left from performing the test.
- 9. Put generator/engine back into service.
- 10. Run engine and verify there are no leaks.

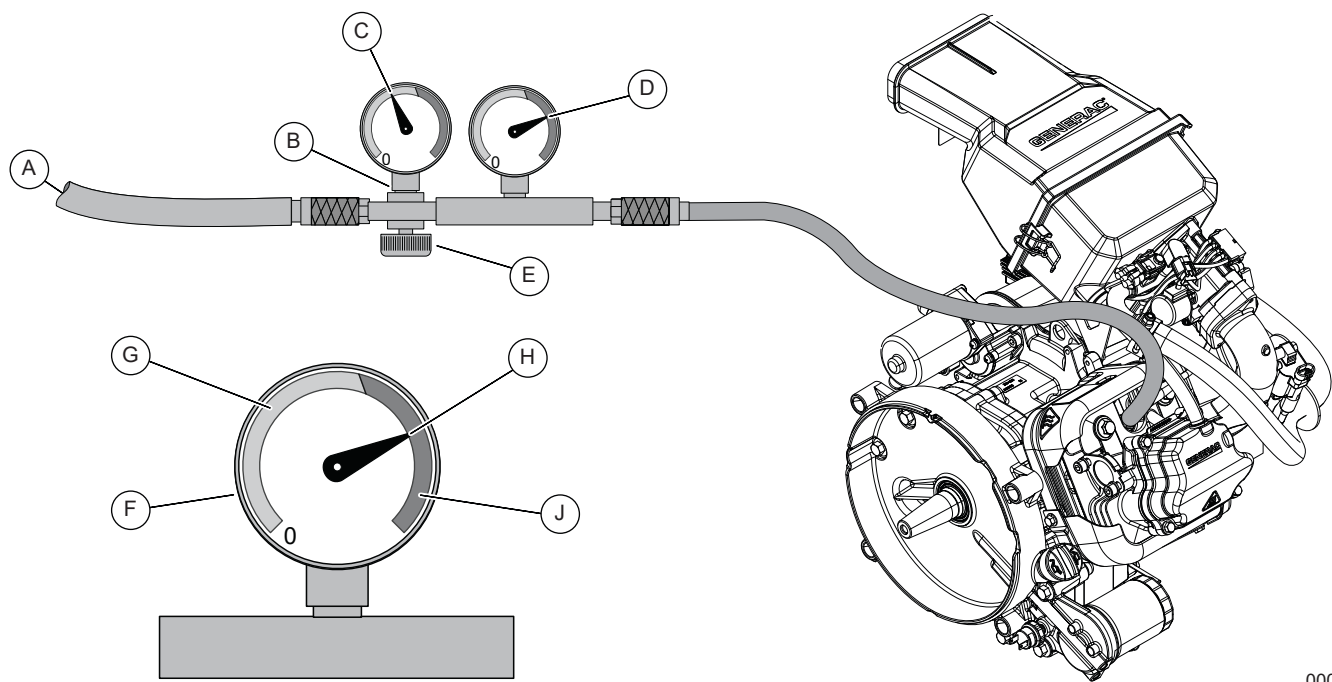
NOTE: See 10000016870 for instructions on checking compression.

Cylinder Leakdown Test

A cylinder leakdown tester (Generac P/N 0F77000SRV) may be used to test the sealing capability of compression components of the cylinder while at TDC of the compression stroke, and quickly identify the problem component.

A cylinder leakdown test will assist in troubleshooting leaking valves or rings.

See [Figure 4-6](#). There are two gauges on the tester. One measures incoming air pressure, and the other measures the percent of cylinder leakage.



00048817

Figure 4-6. Cylinder Leakdown Tester

A	Compressed air inlet
B	Air pressure regulator
C	Inlet gauge pressure set point
D	Outlet gauge pressure
E	Regular adjustment knob
F	Outlet gauge
G	Red range indicates unacceptable air leakage
H	Needle indicates minimal air leakage
J	Green range indicates acceptable air leakage

NOTE: Remove blower housing and cylinder wrappers before proceeding with test. See [Section 6: Cylinder Wrappers and Blower Housing](#).

Cylinder Leakdown Test

Testing Procedure

**⚠ WARNING**

Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury.

(W000111)

Proceed as follows to perform a cylinder leakdown test:

1. Disconnect the ECM.
2. Manually open the throttle plate on the ETC.
3. Connect a remote button starter switch from the battery positive (+) to the starter relay.
4. Align piston at TDC.
5. Install cylinder leakdown tester in the spark plug hole.
6. Pressurize cylinder to 90 psi (621 kPa).
7. Observe cylinder leakage and identify location of air leak.

NOTE: If leaking into intake or exhaust port, inspect valve face and valve seat condition.

Results

- 0–10%—Cylinder compression is acceptable
- 10–30%—Excessive cylinder leakage may exist
- 30–100%—Cylinder requires repair

Factors Affecting Cylinder Condition

- EFI
- Crankcase vacuum
- Ignition timing
- A partially sheared flywheel key will affect ignition timing and engine performance.
- Spark plug
 - A fouled spark plug may indicate fuel injection issues or engine cylinder concerns.
- Leak in spark plug wire
- Head gasket
- Intake manifold
- Valves
- Rings
- Piston
- Cylinder bore

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Section 5: Specifications

Specifications

Model	459CC G26 G-Force Single Engine
Bore	3.622 in (92 mm)
Stroke	2.72 in (69 mm)
Displacement	28.3 in ³ (459 cc)
Oil capacity with oil filter change	1.4 qt (1.32 L)
Oil capacity without oil filter change	1.2 qt (1.14 L)
Common Specifications	
Crank position sensor gap (not adjustable)	0.007 – 0.06 in (0.19 – 1.42 mm)
Stator resistance	0.4 Ohms
Crankshaft end play	0.009 – 0.022 in (0.24 – 0.56 mm)
Spark plug gap	0.020 in (0.50 mm)
Valve lash	Hydraulically adjusted, no manual setting required.

Torque Specifications

P/N	Name	Location Description	Final Torque	Unit
Longblock				
0L1976	SCREW HHFC M8-A.25 X 38LG	Connecting Rod Cap	21.9-26.8	Nm
G076340	SCREW HHFC M6-1.0 X 16 G8.8	Oil Pickup	8-10	Nm
0C3006	Screw HHFC M10-1.5 x 55 G8.8	Gear Cover	43.9-53.7	Nm
0G0926	Screw HHFC M6-1.0 x 16 G8.8	Oil Float Sensor	6.5-7.5	Nm
G045770	Screw HHFC M6-1.0 x 16 G8.8	Pressure Relief Ball Cover	5-6	Nm
Oil System				
0L2917B	LOW OIL PRESS SWITCH 10 PSI	Oil Pressure Switch	*	*
070185E	OIL FILTER	Oil Filter	9.5	Nm
0L6347	3/8-18 NPT	Oil Drain Hose Barb	*	*
Cylinder Head				
10000023678	Screw HWH SEMSM10-1.5 x 100	Cylinder Head Bolts	40.5-49.5	Nm
10000027536	Stud Rocker Ball M6-1.0	Rocker Arm Studs	7.5-9.1	Nm
G080318	Screw HHFC M6-1.0 x 25 FTH G8	Rocker Cover	6.5-7.5	Nm
A0003637864	Spark Plug Generac Gap 0.020"	Cylinder Head Spark Plug	20-30	Nm
0J4967	SCREW HHFC M5-0.8 X 12	Breather Plate	4-4.5	Nm

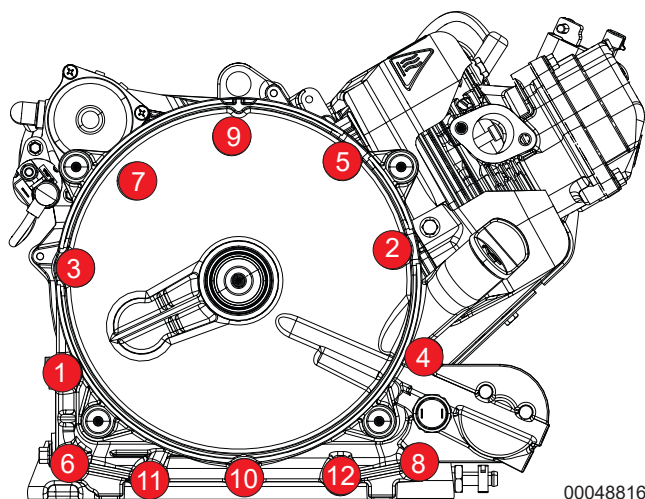
P/N	Name	Location Description	Final Torque	Unit
0D6147	Screw HHFCS M6-1.0 x 10	Cylinder Head Temp Sensor	6.5-7.5	Nm
Flywheel				
0L6110	Screw SHC M6-1.0 x 20	Stator Screws	11-13.6	Nm
0E9507	Screw HHFC M8-1.25 X 14	Stator Ground Screw to Crankcase	25-27.5	Nm
A0000283026	Screw SHC M5-0.8 x 12	CKP & Wire Retention Bracket	6-6.5	Nm
0K8288	Screw HHFCS M8-1.25 x 12	Flywheel Fan	22-27	Nm
G067890	Nut Hex M20-1.5	Flywheel Nut	110-134	Nm
0L4322	Dogpoint Fastener M8-1.25 x 35	Starter Motor	25-30	Nm
G080318	Screw HHFC M6-1.0 x 25	Ignition Coil	10-12	Nm
Intake System				
0G6431A	Screw SHC M6-1.0 x 30 SEMS	Intake Manifold	10-11.5	Nm
0C8563	Screw HHFC M6-1.0 x 12	Airbox Bracket to Crankcase	10-12	Nm
0C9764	Screw HH Hi-Lo ¼-15 x ¾	Airbox Bracket to Airbox	1.13	Nm
A0001231978	Screw HHFC M6-1.0 x 110	Airbox to ETC Mixer	10-12	Nm
0K9104	Screw HHFC M5-0.5 x 10	Low Pressure Sensor on ETC Mixer	7-8	Nm
Trim				
0C8563	Screw HHFC M6-1.0 x 12	Blower Housing to Crankcase	6.5-7.5	Nm
0C9764	Screw HH Hi-Lo ¼-15 x ¾	Wrapper to Blower Housing	1.13	Nm
0C8563	Screw HHFC M6-1.0 x 12	Wrappers to Cylinder Head	10-12	Nm

***NPT Fittings:** As a general rule, pipe fittings with tapered threads should not be assembled to a specific torque. Instead, they should be installed by the number of turns past finger tight. All NPT plugs below 1 in size should be installed to 2-3 turns from finger tight (T.F.F.T.).

Torque Sequences

Proceed with the correct sequence to tighten the gear cover.

(1 — 2 — 3 — 4 — 5 — 6 — 7 — 8 — 9 — 10 — 11)

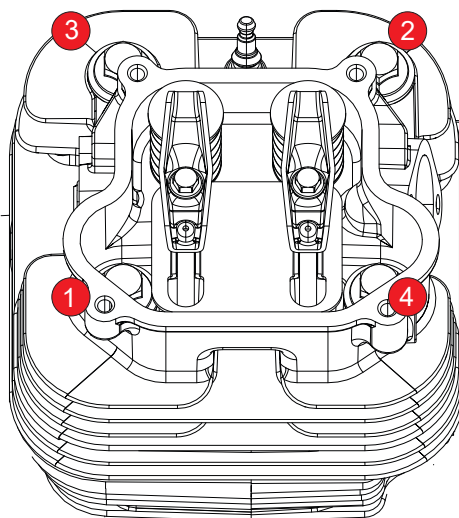


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Figure 5-1. Gear Cover Torque Sequence

Proceed with the correct sequence to tighten the cylinder head.

(1 — 2 — 3 — 4)



00048815

Figure 5-2. Cylinder Head Torque Sequence

Component Weights

Part Description	Part Number	Weight (kg)[lbs]
Cylinder Head	A0002106973	(1.99 kg) [4.39 lbs]
Complete Engine Assembly (Dry)	A0000682951	(45.63 kg) [100.60 lbs]

Approved Sealants/Lubricants

- Thread sealants (used for threads which require sealant):
 - Permabond LH150 (or equivalent)
 - Permatex® High Temperature Thread Sealant (or equivalent)
 - LOCTITE® 567™ (or equivalent)
- RTV sealants (used for gasket between crankcase and gear cover):
 - Permatex® Ultra Black® Gasket Maker (or equivalent)
 - Loctite SI 593™ Black (or equivalent)
- Assembly lube (used when applying parts which require lubricant for installation):
 - Lube Tech T24872 (or equivalent)

Section 6: Engine Disassembly

Disassembling Long Block

Proceed as follows to disassemble the long block:

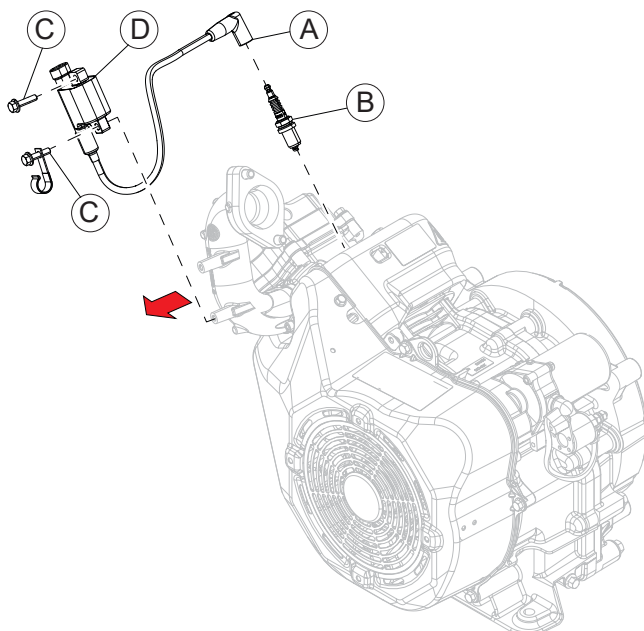
1. Drain oil. Dispose of used engine oil in accordance with state and local codes.
2. Disconnect wire harness, unplug connector cables, and disconnect fuel hose.
3. Disconnect intake and exhaust systems from equipment.
4. Remove engine from equipment.

Long Block Disassembly

Removing Ignition Coil and Spark Plug

Proceed as follows to remove ignition coil:

1. See [Figure 6-1](#). Remove spark plug wire (A) from spark plug (B).
2. Remove two fasteners (C) and the ignition coil (D) from mounting position on intake manifold.



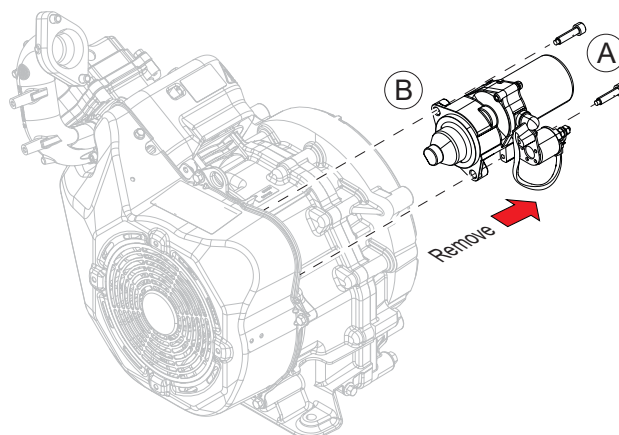
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Figure 6-1. Remove Ignition Coil and Spark Plug

Removing Spark Plug and Starter

Proceed as follows to remove spark plug and starter:

1. See [Figure 6-1](#). Remove spark plug (B) from cylinder head.
2. See [Figure 6-2](#). Remove two fasteners (A) and starter motor (B) from the crankcase.



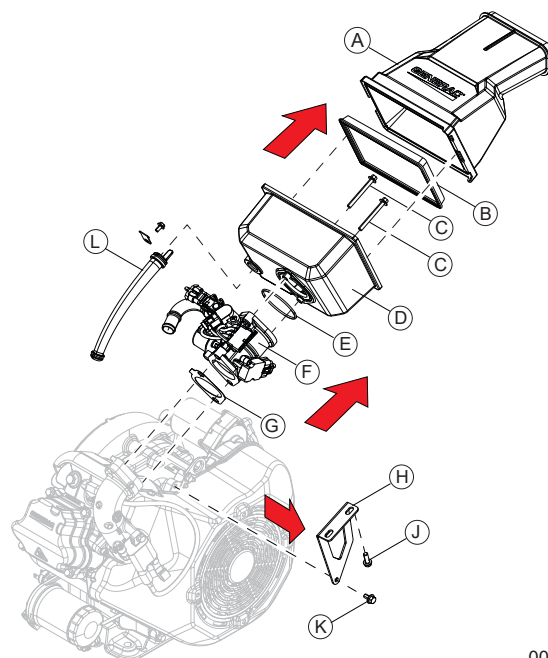
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Figure 6-2. Remove Starter

Removing Air Box

Proceed as follows to remove air box:

1. See [Figure 6-3](#). Release clasp securing air box cover (A) to air box base (D).



00049494

Figure 6-3. Remove Air Box and Mixer Assembly

2. Tip air box cover away from hatch to release air box cover from connecting tabs on opposite side.
3. Remove air filter media (B) from air box base (D).
4. Remove the two bolts (C) from inside air box base attaching air box to mixer.

NOTE: Mixer is now free from its bolted mounting and caution should be taken to not damage mixer when removing the rest of the air box assembly.

5. Remove air box base mounting bolts (K) from cylinder head and gently dislodge breather hose (G) from air box. Do not damage breather hose when removing from air box.
6. Lift air box base (D) assembly from engine.

Removing Breather Hose From Rocker Cover

See [Figure 6-3](#). Gently remove breather hose (L) from rocker cover.

Removing Mixer Assembly

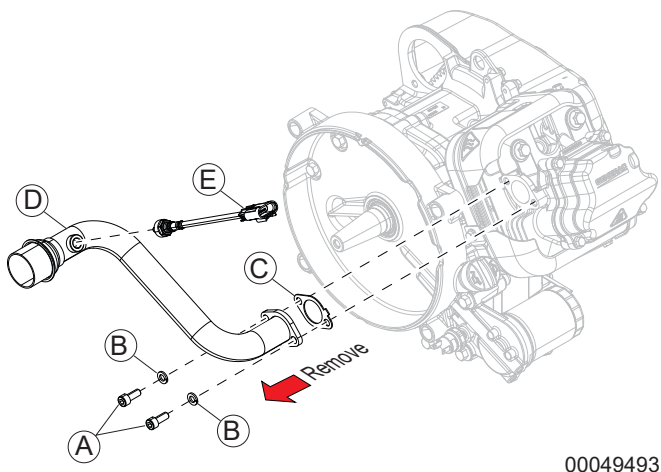
Proceed as follows to remove mixer assembly:

1. See [Figure 6-3](#). Remove O-ring (E) from groove on face of mixer (F) where it mates the air box bottom.
NOTE: With the air box removed from the engine, there will no longer be bolts attaching mixer to engine. Do not damage mixer or O-ring upon disassembly.
2. Carefully remove mixer.
3. Remove mixer gasket (G) from intake manifold. Do not damage mixer gasket.

Removing Intake and Exhaust Components

Proceed as follows to remove intake and exhaust components:

1. See [Figure 6-4](#). Remove oxygen sensor (E) from the exhaust manifold. Replace if needed.

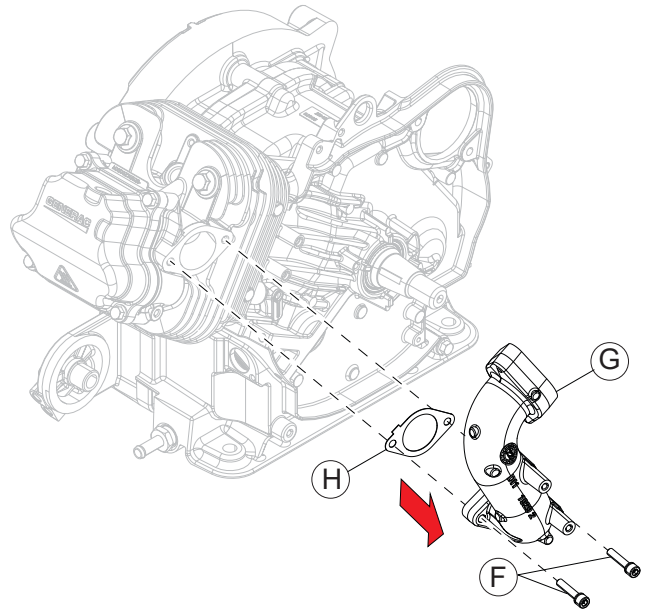


00049493

Figure 6-4. Remove Exhaust and Intake Manifold

2. Remove two exhaust manifold bolts (A) and washers (B).
3. Remove exhaust manifold (D).
4. Discard old gaskets (C). Use new gaskets when assembling.

5. See [Figure 6-5](#). Remove two bolts (F) and intake manifold (G) from the cylinder head. Discard intake manifold gasket (H).



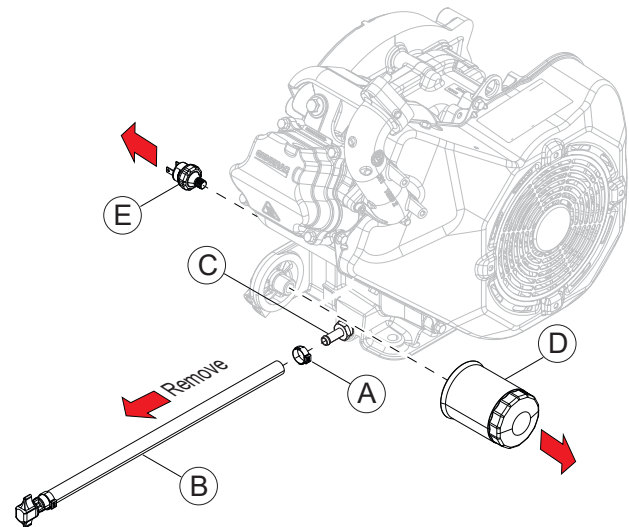
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Figure 6-5. Manifold Gaskets

Removing Oil Drain Line

Proceed as follows to remove oil drain line:

1. See [Figure 6-6](#). Cut off the crimped ear of the stepless ear clamp (A) to remove it from the oil drain hose assembly (B).



00049491

Figure 6-6. Remove Oil Drain Hose

NOTE: Factory installed stepless ear clamps are single use and must be replaced after removal.

2. Remove oil drain hose assembly (B) from barbed fitting (C).
3. Remove barbed fitting from crankcase.

Removing Oil Filter and Low Oil Pressure Switch

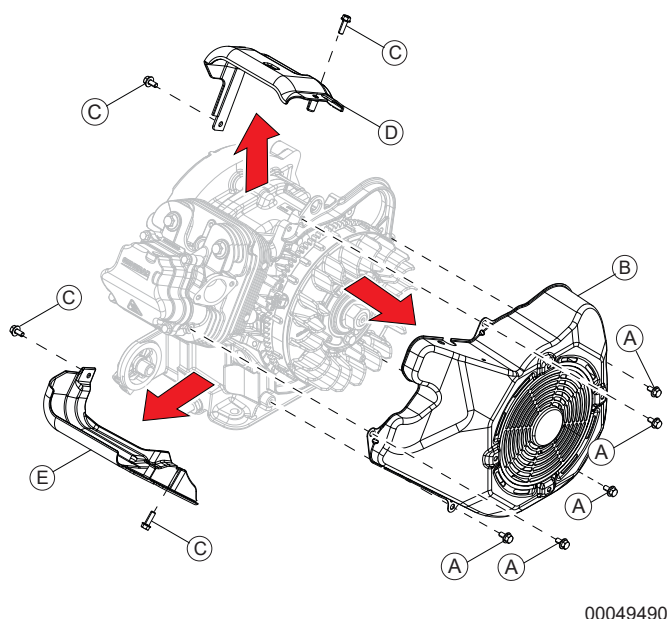
See [Figure 6-6](#). Remove oil filter (D) and low oil pressure switch (E) from oil filter mount.

Removing Cylinder Wrappers and Blower Housing

Proceed as follows to remove cylinder wrappers and blower housing:

1. See [Figure 6-7](#). Remove five fasteners (A) and blower housing (B).

NOTE: Disengage the plastic tab, located on the bottom of the housing, before removal.



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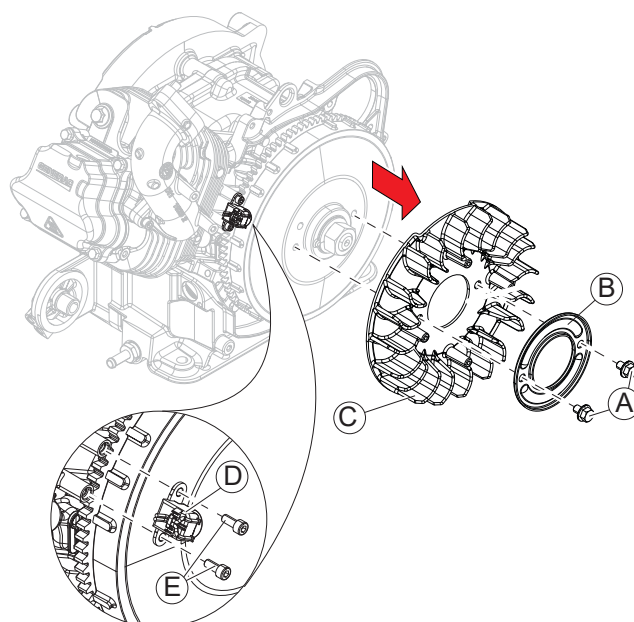
Figure 6-7. Remove Blower Housing and Cylinder Wrappers

2. Remove four fasteners (C) and side wrappers (D, E).

Removing Flywheel

Proceed as follows to remove flywheel:

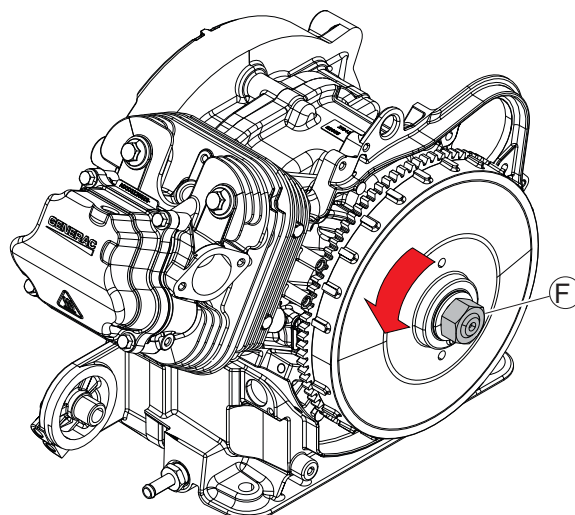
1. See [Figure 6-8](#). Remove two fasteners (A), fan plate (B), and fan (C) from flywheel.



00048814

Figure 6-8. Remove Fan

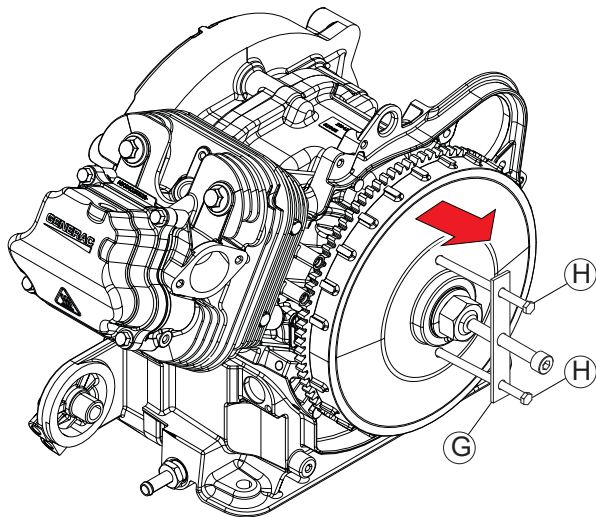
2. Remove bolts (E) securing crankshaft position sensor (D). Remove crankshaft position sensor from engine.
3. See [Figure 6-9](#). Loosen flywheel nut (F) until it is flush with end of crankshaft threads.



00049066

Figure 6-9. Loosen Flywheel Nut

4. See [Figure 6-10](#). Install flywheel puller (G) using two M8 x 1.25 mm flywheel puller bolts (H).



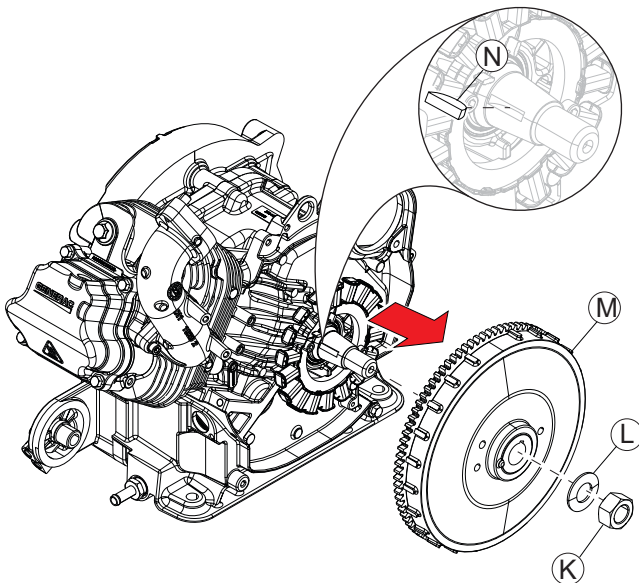
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Figure 6-10. Install Flywheel Puller

5. Tighten flywheel puller bolts until flywheel loosens.
6. Remove flywheel puller.

IMPORTANT NOTE: Do NOT over tighten flywheel puller bolts. DO NOT strike flywheel with a hard object or a metal tool. Doing so may cause flywheel to shatter in operation. Always use approved flywheel removal tools.

7. See [Figure 6-11](#). Remove nut (K), washer (L), flywheel (M), and woodruff key (N) from crankshaft.



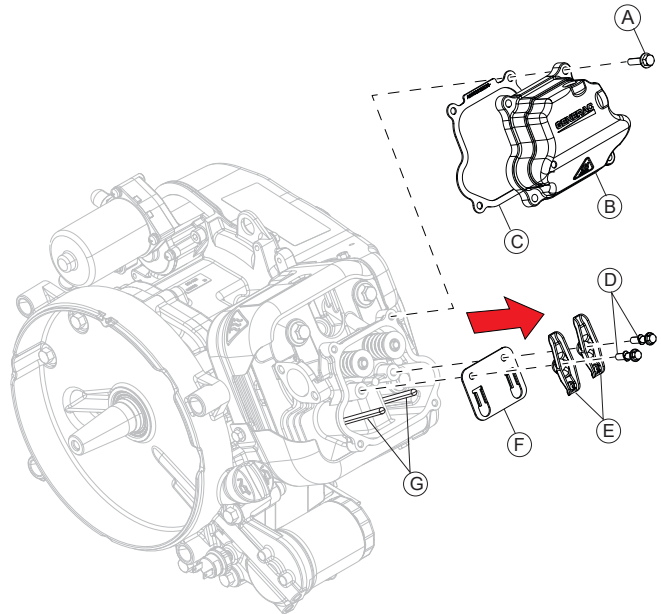
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Figure 6-11. Remove Flywheel

Removing Rocker Covers

Proceed as follows to remove rocker covers:

1. See [Figure 6-12](#). Remove the four fasteners (A) and rocker cover (B) from cylinder head.



00049489

Figure 6-12. Remove Rocker Cover

2. Remove and discard gasket (C).

Removing Pushrods and Rocker Arms

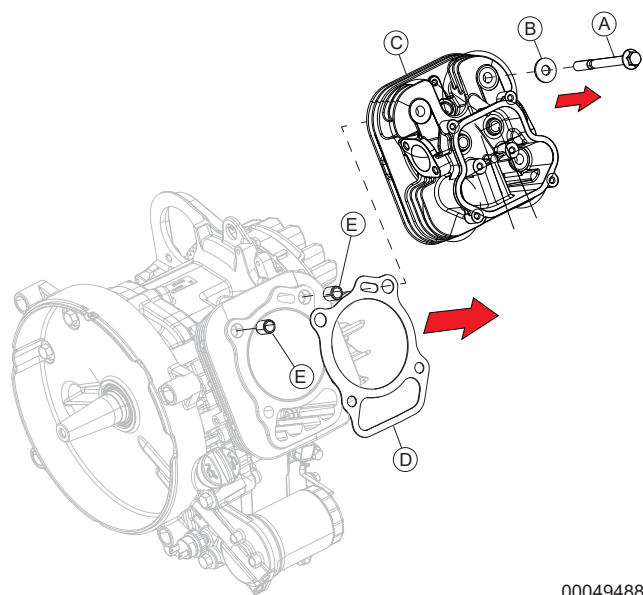
Proceed as follows to remove pushrods and rocker arms:

1. See [Figure 6-12](#). Remove rocker arm pivot studs (D), rocker arms (E), and pushrod guide plate (F) from cylinder head.
2. Remove the two pushrods (G).

Removing Cylinder Head

Proceed as follows to remove cylinder heads:

1. See [Figure 6-13](#). Remove the four cylinder head bolts (A), washers (B), cylinder head assembly (C), and head gasket (D) from engine long block. Discard head gasket.



00049488

Figure 6-13. Remove Cylinder Head Assembly

2. Remove the cylinder head bolt dowels (E) from cylinder head assembly.

Disassembling Cylinder Head

⚠ WARNING

Personal Injury. Always wear all necessary protective equipment and use recommended tools while performing the following procedure. Failure to do so could result in serious injury.

(W000556)

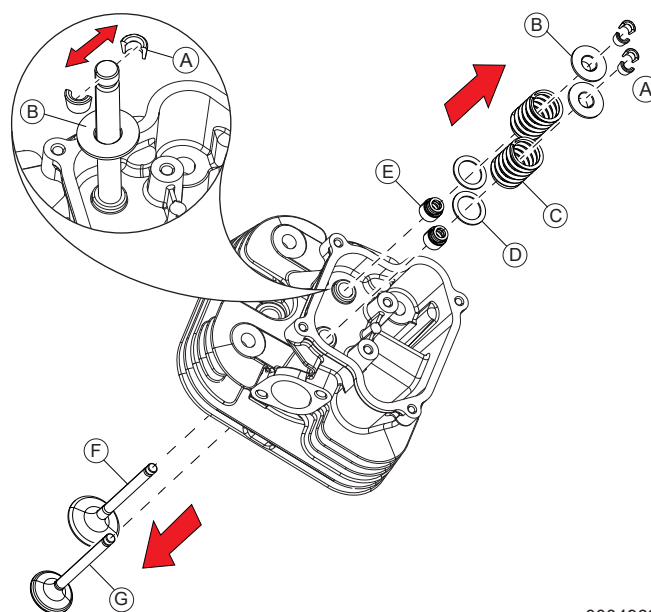
⚠ WARNING

Personal Injury. Properly compress valve springs using a valve spring compressor. Failure to do so could cause the spring to eject, and could result in serious injury.

(W000193)

Proceed as follows to disassemble cylinder head:

1. See [Figure 6-14](#). Use a valve spring compressor to compress valve spring retainer (B) and valve spring (C) sufficiently to remove the two valve spring keepers (A) from valve stem.



00049625

Figure 6-14. Disassembling Cylinder Head

2. Remove valve spring retainer (B) and valve spring (C).

IMPORTANT NOTE: Spring compressor is under a large amount of tension. Release tension carefully.

NOTE: Valve will be loose in cylinder head. Do not damage valve by allowing valve to drop out of cylinder head.

3. Remove the valve spring seat washer (D).
4. Remove intake valve (F) from combustion chamber side. Do not damage valve stem seal.
5. Remove valve stem seal (E) from cylinder head intake guide.
6. Repeat steps 1 – 4 to remove exhaust valve (G) from cylinder head assembly.

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Section 7: Ignition

Inspecting Ignition

Inspect ignition system for correct function if engine runs but misfires during operation. Proceed as follows to inspect ignition system:

1. See [Figure 7-1](#). Install a spark tester (A) (Generac part number 0C5969) between spark plug lead (B) and spark plug (C).

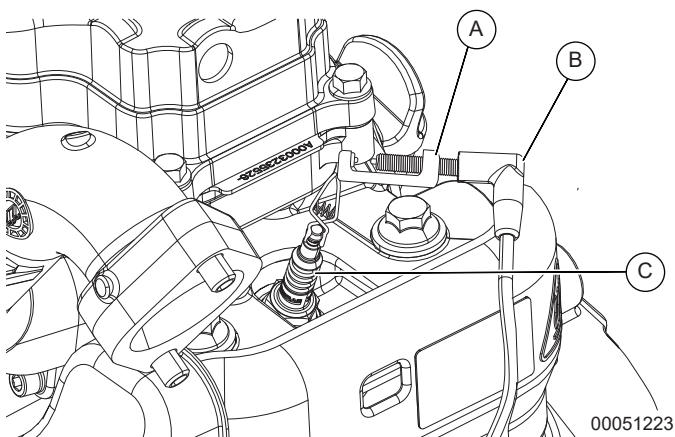


Figure 7-1. Spark Tester on Engine (Typical)

2. Crank engine and verify a steady spark is visible in the window of the tester. A lack of spark or misfire will be readily apparent when engine is running, indicating further investigation of the ignition system is required.
3. If spark is steady but engine still misfires or will not run, inspect for a fouled spark plug. Verify fuel is present in the fuel system and power is being supplied to the fuel shutoff solenoid.
4. If there is no spark, connect spark tester to engine block (ground) and crank engine. If spark is present, spark plug is damaged.

Removing Ignition Coil

Proceed as follows to remove the ignition coil for testing:

1. See [Figure 7-2](#). Remove spark plug wire (A) from spark plug (B).

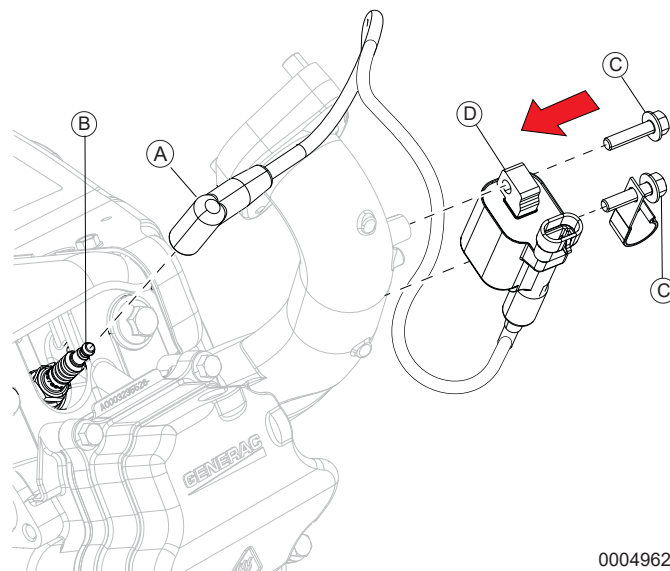


Figure 7-2. Remove Spark Plug Wires and Ignition Coils from Intake Manifold

2. Remove bolts (C) retaining ignition coil (D) to intake manifold and remove ignition coil.
3. Inspect coil wire lead to verify it has not worn through allowing wiring to ground to backing plate. If contact is identified, repair/replace wire and check for spark.

Testing Ignition Coil

Proceed as follows to test ignition coil if ignition inspection indicates one or both of the ignition coils are not delivering a strong consistent spark:

1. Disconnect electrical connection and primary wire from the coil.

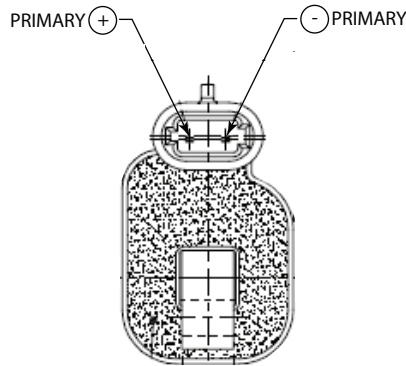
NOTE: It may be necessary to remove the coil from the mounting position.

2. Set digital multimeter (DMM) to measure resistance.
3. See [Figure 7-3](#). Attach positive (+) test lead to primary wire terminal 451 to measure resistance across coil.



Figure 7-3. Ignition Coil Wiring Diagram

4. Attach negative (-) test lead to wire terminal 14C to measure resistance.
5. See [Figure 7-4](#). Measure resistance across coil. Resistance on primary side should read 1 ohms $\pm$ 15%.



00052009

Figure 7-4. Ignition Coil

6. Replace ignition coil if coil is out of spec.

Inspecting Wire Harness

If ignition does not have spark, disconnect bullet connection from ignition grounding stud on engine and inspect again. If ignition now has spark, inspect wire harness for pinched wires, broken insulation, or (if equipped) faulty diode. If there is no spark, proceed to [Removing Ignition Coil](#) section.

Flywheel

The flywheel is a weighted disc which helps keep the crankshaft rotating between power strokes and provides a reference for the crank position sensor to set the ignition timing.

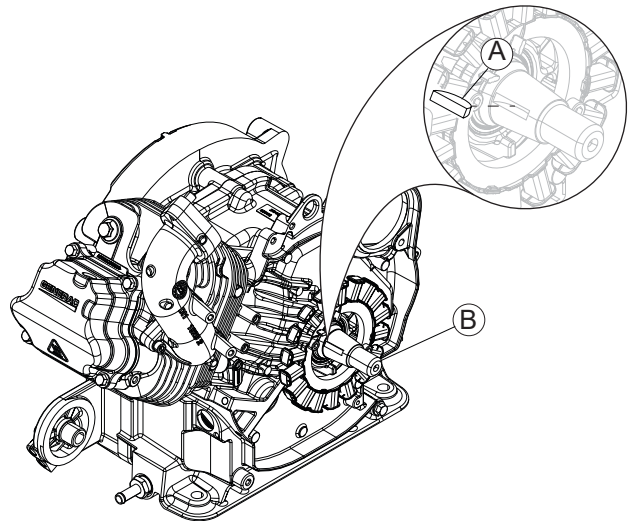
Removing Flywheel and Fan

See [Fan and Flywheel](#) removal section to correctly remove fan and flywheel from engine.

Installing Flywheel and Fan

Proceed as follows to install flywheel and fan:

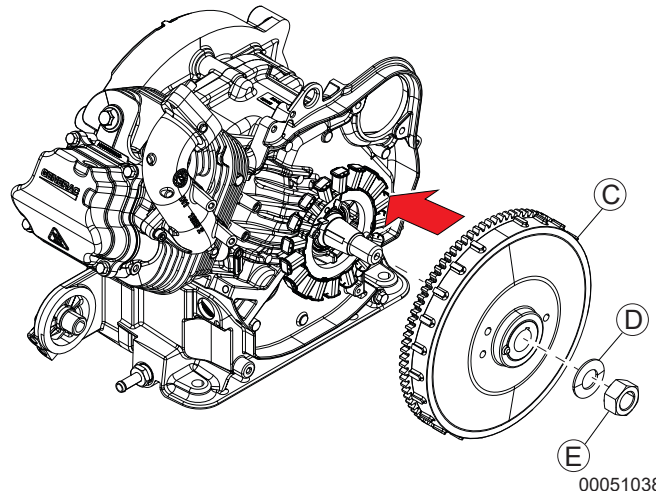
1. See [Figure 7-5](#). Install woodruff key (A) into slot in crankshaft (B).



00051039

Figure 7-5. Install Woodruff Key

2. Verify both crankshaft and flywheel tapers are clean and free of oil and debris.
3. See [Figure 7-6](#). Align keyway and install flywheel (C) on crankshaft.

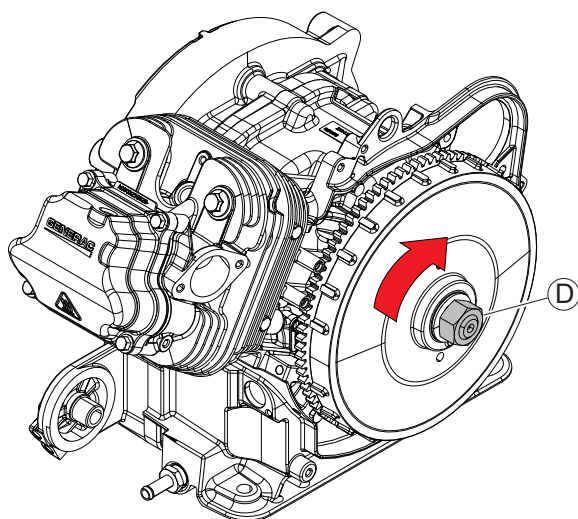


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Figure 7-6. Install Flywheel, Washer, and Nut

NOTE: Flywheel nut (E) and flywheel washer (D) are part of A0007674900 KIT-FAN.

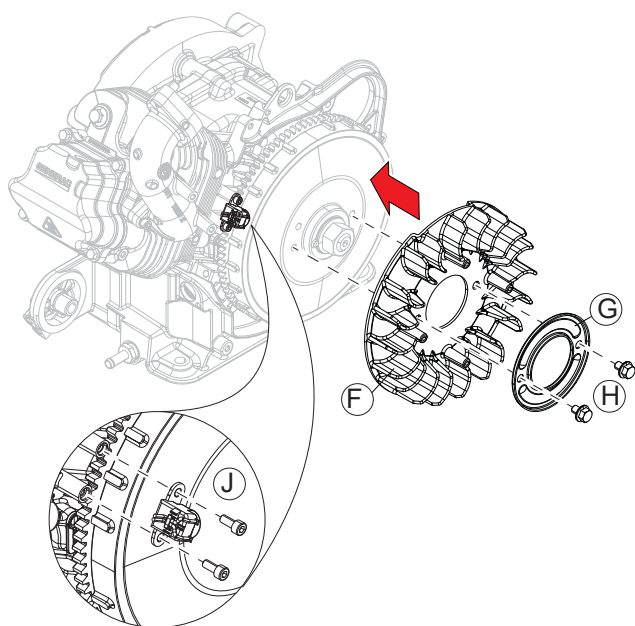
4. Attach flywheel to crankshaft using flywheel washer and flywheel nut.
5. See [Figure 7-7](#). Tighten flywheel nut (D) to 81.13 - 98.8 ft-lbs (110 – 134 Nm).



00049622

Figure 7-7. Tighten Flywheel Nut

6. See [Figure 7-8](#). Install fan (F) and fan plate (G) to flywheel using two fasteners (H).



00049621

Figure 7-8. Install Flywheel Fan

NOTE: Fan, fan plate, and bolts are part of A0007674900 KIT-FAN.

7. Tighten bolts to 16–20 ft-lb (22–27 Nm).

Installing Crankshaft Position Sensor

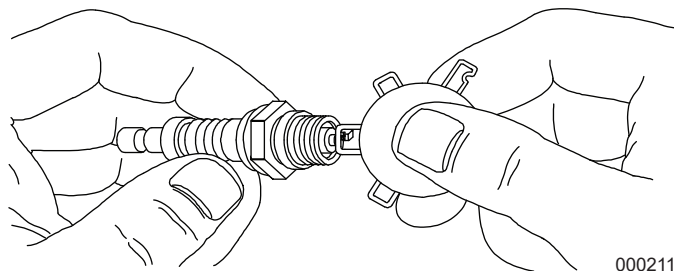
Proceed as follows to install crankshaft position sensor:

1. See [Figure 7-8](#). Install crankshaft position sensor (J) using fasteners.
2. Tighten fasteners to 18.4–20.3 ft-lb (25–27.5 Nm).

Checking Spark Plug Gap

Spark plug should be cleaned and re-gapped periodically. Spark plug should be replaced every 200 hours of operation or once annually, whichever comes first.

See [Figure 7-9](#). Set spark plug gap to 0.020 in (0.50 mm) using a wire-type feeler gauge.



000211

Figure 7-9. Check Spark Plug Gap

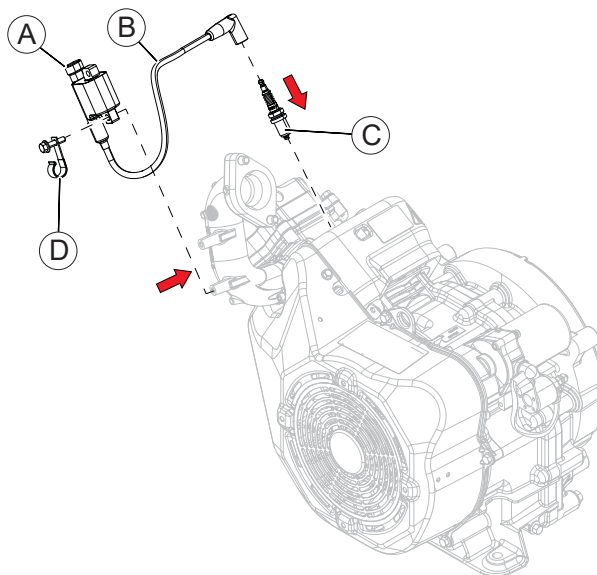
NOTE: See owner's manual for the specific product for correct spark plug replacement. Always use original Generac replacement parts.

Installing Ignition Coil

Ignition coil mounts to the intake manifold and only can be installed after installation of the intake manifold.

Proceed as follows to install ignition coil:

1. See [Figure 7-10](#). Install ignition coil (A) with spark plug lead (B) pointing downward.



00049620

Figure 7-10. Install Ignition Coil and Spark Plug Wire

2. Fasten coil to intake manifold using two fasteners and tighten to 88.5–106.2 in-lb (10–12 Nm).

NOTE: Attach oil drain hose hanging clip to bottom fastener (D) when attaching coil.

3. Install spark lead (B) to spark plug (C).

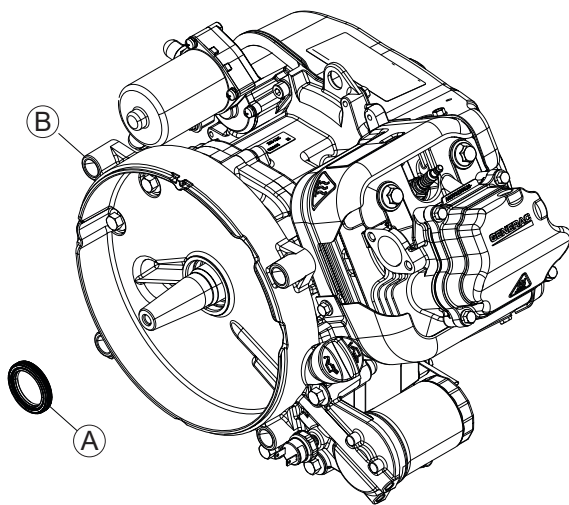
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Section 8: Sub-Assemblies

Gear Cover Sub-Assembly

Proceed as follows to install oil seal and check torque on gear cover assembly:

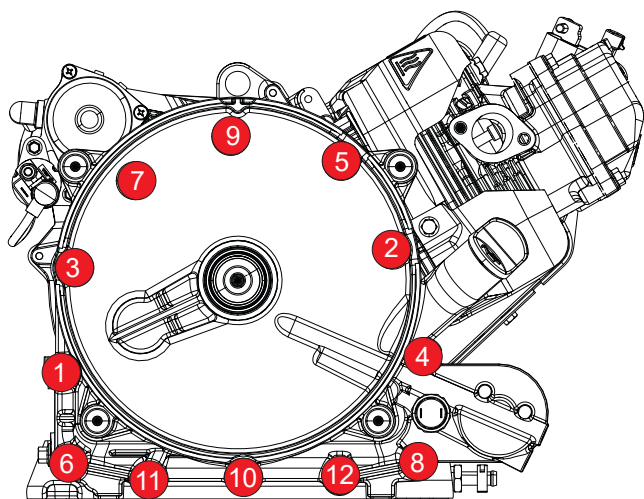
1. See [Figure 8-1](#). Press the crankshaft PTO oil seal (A) into the gear cover assembly (B) until seal is fully seated against bottom of bore.



00051014

Figure 8-1. Install Oil Seal Into Gear Cover

2. See [Figure 8-2](#). Check gear cover torque using the correct sequence shown. Final torque on gear cover is 32.4-39.6 ft-lb (43.9-53.7 Nm).



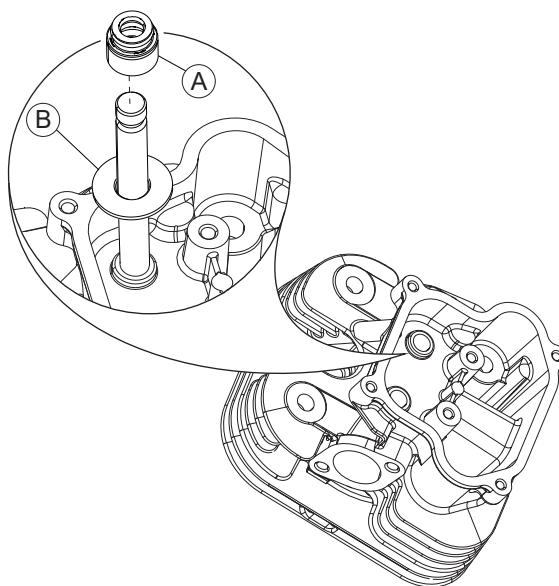
00051013

Figure 8-2. Gear Cover Torque Sequence

Cylinder Head Sub-Assembly

Proceed as follows to complete the cylinder head sub-assembly:

1. See [Figure 8-3](#). Press valve stem seal (A) onto cylinder head intake and exhaust guide.

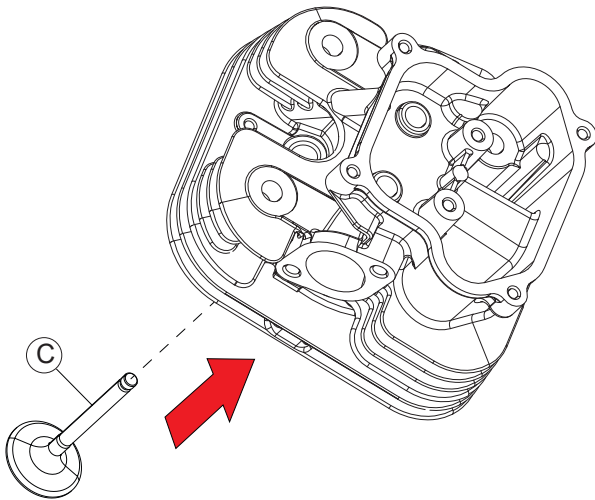


00051012

Figure 8-3. Install Valve Stem Seal onto Cylinder Head

2. Install valve spring washer (B) over valve guide seal.
3. See [Figure 8-4](#). Apply assembly lube to the tip of intake valve (C) and insert intake valve into intake valve guide from combustion chamber side.

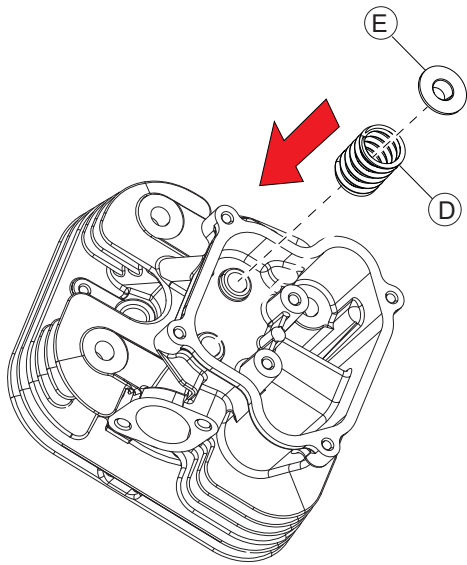
NOTE: Take special care to not damage seal upon insertion.



00051011

Figure 8-4. Install Intake Valve into Cylinder Head

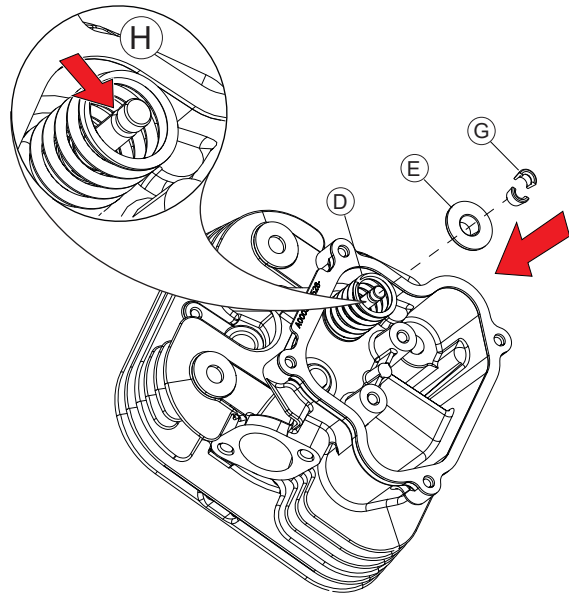
4. See [Figure 8-5](#). Install valve spring (D) and spring retainer (E) over valve stem and onto cylinder head assembly.



00051010

Figure 8-5. Install Valve Spring Assembly onto Valve

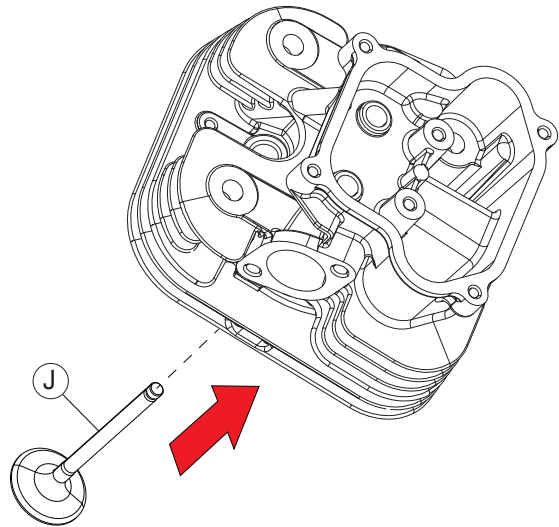
5. See [Figure 8-6](#). Use a valve spring compressor to compress the valve spring (D) sufficiently enough to install the two valve spring keepers (G). Verify valve spring keepers are locked into the slot of the valve stem (H).



00051009

Figure 8-6. Install Valve Spring Keepers onto Valve

6. Release valve spring compressor. Verify valve spring keepers are firmly locked into slot on valve stem.
7. See [Figure 8-7](#). Repeat steps 1– 5 to install exhaust valve (J) into cylinder head assembly.



00051008

Figure 8-7. Install Exhaust Valve into Cylinder Head

Section 9: Cylinder Head and Valves

Inspecting and Repairing Cylinder Head

Proceed as follows to inspect and repair cylinder head:

1. See [Figure 9-1](#). Inspect condition of cylinder head. Verify all gasket material is removed from surfaces before inspecting.
 - a. Inspect cylinder head for cracks or damage.
 - b. Use a surface plate or straightedge and inspect cylinder head mounting surface for distortion.

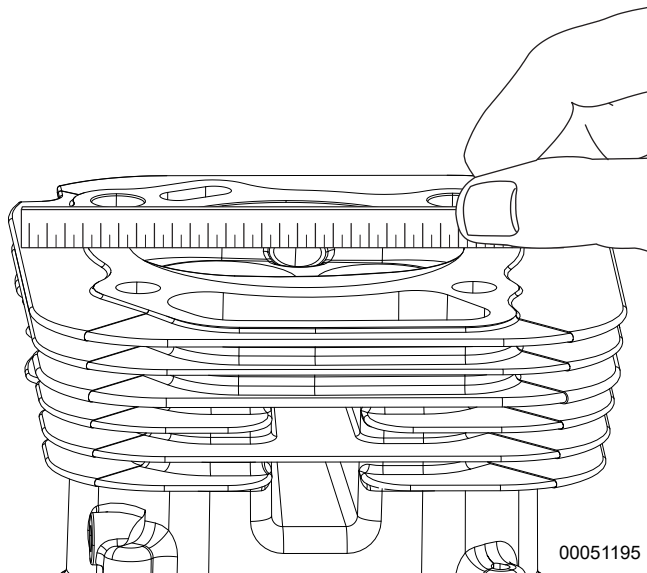
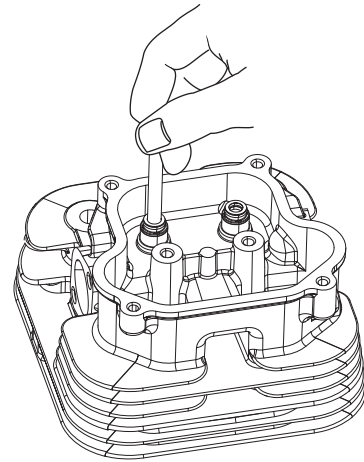


Figure 9-1. Check Cylinder Head Distortion

Cylinder head must be replaced if mounting surfaces are distorted more than 0.004 in (0.1 mm).

NOTE: Resurfacing cylinder head mounting surfaces is not recommended.

2. See [Figure 9-2](#). Clean valve guides and measure inner diameter using a split ball bore gauge. Replace head if either valve guide measures 0.278 in (7.06 mm) or more.



00051216

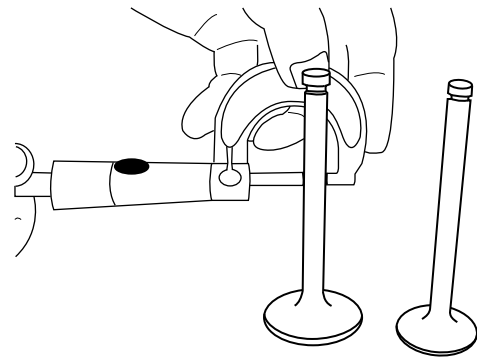
Figure 9-2. Measure Valve Guide Inner Diameter

NOTE: Valve guides are not replaceable.

Measuring Valves and Springs

Proceed as follows to measure valve stem diameter and valve springs:

1. See [Figure 9-3](#). Measure valve stem diameter at 1.600 in (40 mm) from end of valve.

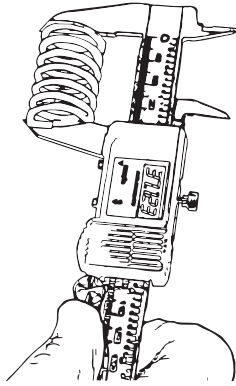


002252

Figure 9-3. Measure Valve Stem Diameter

Replace valves if stem diameter is less than 0.272 in (6.9 mm), or if total clearance between valve stem and valve guide exceeds 0.0047 in (0.12 mm).

2. See [Figure 9-4](#). Check valve springs for free length. Replace valve springs if free length is less than 1.437 in (36.5 mm).



004442

Figure 9-4. Check Valve Springs

Gasket Removal

Verify sealing surface is completely clean and free of old gasket material before installing a new gasket. Residual gasket material or dirt can prevent the new gasket from sealing completely. An incorrectly sealed gasket may cause the engine to leak oil, lose compression, or overheat.

It is critical to completely remove old gasket material without damaging sealing surfaces. Aluminum is very soft and easy to scratch and gouge. A new gasket may not seal completely if the underlying surface is damaged. Generac does not recommend using a wire brush or abrasive sanding to remove gaskets. Use a metal gasket scraper specifically designed for this purpose. Scrape gasket at a shallow angle—almost parallel to the sealing surface. Wipe sealing surface with a clean, dry, lint-free rag after removing gasket residue.

If the gasket is difficult to scrape off, spray it with an aerosol gasket removal compound. Allow the compound to soak into the gasket for the length of time recommended by the manufacturer (typically 10– 20 minutes), then use the gasket scraper again.

NOTE: Some gasket removal compounds contain chemicals damaging to plastics. Read and follow the directions on the product label before use.

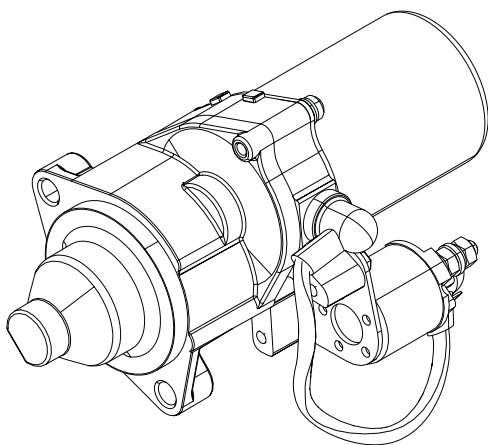
Assemble Cylinder Head

See Cylinder Head Assembly in [Section 8: Cylinder Head Sub-Assembly](#).

Section 10: Electric Starter

General Information

See [Figure 10-1](#). The starter motor utilizes a solenoid, similar to an automobile starter, to assist in pinion gear engagement. When the starter motor is activated, the pinion gear engages a ring gear attached to the engine flywheel and cranks the engine.



021329

Figure 10-1. Starter Motor

Troubleshooting

IMPORTANT NOTE: Do not rotate engine with electric starter while spark plugs are removed, unless using a compression gauge. Arcing at the spark plug ends may ignite fuel vapor exiting the spark plug hole.

NOTE: If a starting problem is encountered, inspect the engine thoroughly to eliminate it as the cause of starting difficulty. Check engine for free rotation by removing spark plugs and flywheel guard and turning the crankshaft with a socket and ratchet or breaker bar. Verify it rotates freely.

Starter Troubleshooting

Engine cranks slowly:

- Additional drag on engine, affecting starter performance (see note above).
- Discharged battery.
- Faulty electrical connection (battery circuit).
- Dirty or worn starter motor commutator, bearing, weak magnets, worn brushes, etc.
- Incorrect oil viscosity for temperature expected.
- Battery leads too long or wire too small.
- Underpowered battery.

Engine will not crank if:

- Faulty safety interlocks.
- Discharged battery.
- Faulty electrical connections.
- Faulty starter motor switch (open circuit).
- Open circuit in starter motor.
- Brushes sticking, etc.
- Faulty solenoid.

Starter motor spins but does not crank engine:

- Sticking pinion gear due to dirt.
- Damaged pinion or ring gear.
- Damaged or faulty battery.
- Incorrect rotation due to reversed motor polarity.
 - All motors rotate counterclockwise viewed from pinion gear.
- Damaged solenoid.

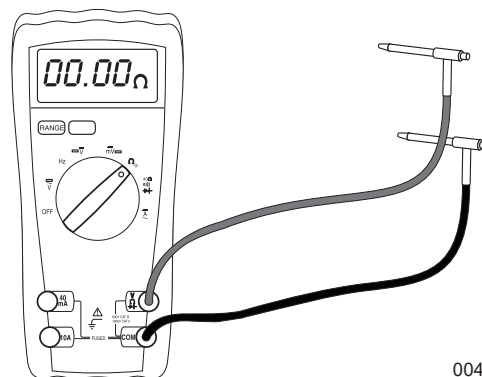
Starter motor spins but will not stop:

- Faulty starter switch.
- Solenoid stuck engaged.

Test Equipment

Digital Multimeter (DMM)

See [Figure 10-2](#). A DMM may be used to read volts, ohms, amperes, and test diodes (rectifiers).



004298

Figure 10-2. Digital Multimeter (DMM)

NOTE: The DMM is equipped with two fuses to prevent DMM damage if input limits are exceeded. Inspect fuses if the DMM displays a reading of 0.00 when testing DC output. See DMM operators' manual for more information.

Testing the Starter Motor

Checking the Pinion

The pinion gear should rise and engage the ring gear when the starter motor is activated. If the pinion gear does not rise normally, inspect the large spiral threads of the sleeve and pinion for binding or sticking.

Tools for Starter Performance Test

The following equipment may be used to complete a performance test of the starter motor:

- a digital multimeter (DMM)
- a tachometer capable of reading up to 1,500 rpm
- a fully charged 12 volt battery

Testing Starter Performance

Proceed as follows to test starter performance:

1. See [Figure 10-3](#). Set voltmeter (A) to read DC amps.

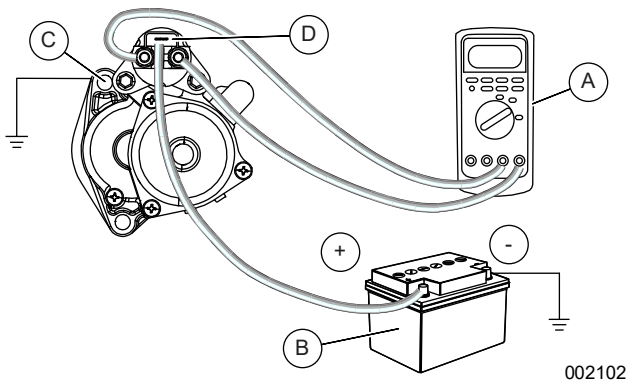


Figure 10-3. Starter Performance Test

2. Connect voltmeter as shown.
3. Ground battery (B) and starter casting (C).
4. Connect one end of jumper cable to starter spade terminal (D).
5. Touch free end of jumper cable to positive battery lead to close the solenoid. An audible click should be heard as the solenoid switch “closes.”

Starter motor performance specifications at 12 volts DC:

- Minimum motor rpm = 800
- Maximum amps = 9

Replace starter motor if it does not perform satisfactorily.

Verifying Starter Voltage Drop

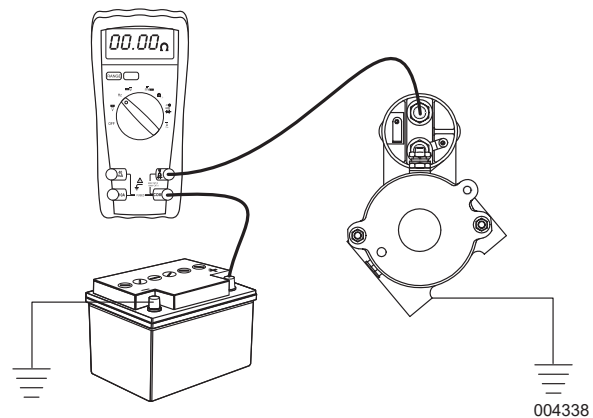
Verify battery is at least 70% charged (12.5 V or higher) before starting this test. Verify a correctly sized battery is being used. See engine manufacturer's operator's manual for battery size recommendations.

Measure battery voltage across the battery posts using a DMM set to the 20V scale with engine and all other accessories turned off. Do not measure across battery wire terminal ends. Voltage should read 12.5 V or higher.

NOTE: Do not crank engine longer than 15 seconds during any of the following tests.

Proceed as follows to verify starter voltage drop:

1. To prevent the engine from starting: disconnect the ECM, manually open throttle plate on the ETC, and connect a remote button starter switch from the battery positive (+) to the starter relay.
2. Place DMM positive (+) lead on positive battery terminal and negative (-) lead on negative battery terminal. Crank engine and observe DMM reading.
3. Place DMM positive (+) lead on starter battery terminal stud and negative (-) lead on starter housing. Crank engine and observe DMM reading.
4. Compare the two voltage readings. If readings are the same (or vary within one volt or less), there is no excessive voltage drop on the positive side. If a voltage drop of greater than one volt exists, proceed to step 5.
5. See [Figure 10-4](#). Set DMM on the DC 2V scale. Place DMM negative (-) lead on positive (+) battery terminal and DMM positive (+) lead on starter positive (+) terminal stud. With engine still disabled, crank engine and observe DMM reading. There should be less than a 0.2 V drop. If drop is greater than 0.2 V, test each wire and terminal connection. No more than 0.2 volts should be observed at each connection.



6. Test negative (-) side of system. Place DMM positive (+) lead on starter case and DMM negative (-) lead on negative (-) battery terminal.
7. Crank engine and observe the reading. Voltage should be less than 0.2 V.
8. If voltage reads greater than 0.2 volts, test all wire connections on the starter negative (-) side. There should be no readings higher than 0.2 volts.

Testing Starter Amp Draw

⚠ CAUTION

Equipment Damage. Do not continuously crank engine for more than ten seconds. Doing so will lead to overdischarge of batteries and starter seizure. (C000230)

This test requires a special clamp-on tool called an ammeter. An engine repair shop should be able to perform this test.

Proceed as follows to disable engine from starting for testing:

1. Disconnect ECM.
2. Manually open throttle plate on the ETC.
3. Connect a remote button starter switch from the battery positive (+) to the starter relay.

Proceed as follows to test starter amp draw:

1. Place DC ammeter clamp on positive (+) cable running from the positive (+) battery terminal to starter post.
2. Ground ignition and crank engine for no longer than 10 seconds. Starter amp draw should be less than 100 amps during cranking.

Excessive draw could be caused by the following:

- Faulty starter. To verify, remove starter from engine and bench test prior to replacing. Perform an amp draw test on the bench. Starter should not exceed 62 amps with a minimum pinion speed of 3,250 rpm.
- Engine oil viscosity too heavy (see engine manufacturer's operators manual).
- Some equipment manufacturers may use a constant drive pump. Constant drive pump oil may be too heavy or belt tension may be too tight.
- Internal engine problem. Turn engine over with a socket and ratchet to verify it rotates easily.
- Battery cable loose or incorrectly sized.
- Battery terminals corroded.
- Battery CCA too low.

Battery Information

⚠ WARNING



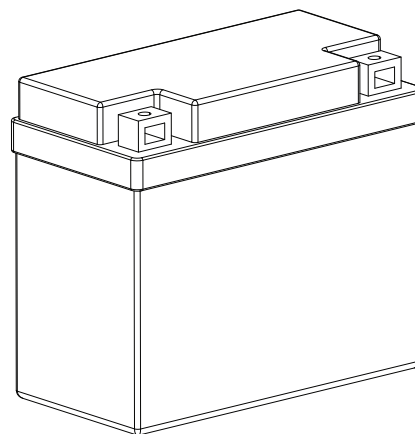
Explosion. Batteries emit explosive gases. Always disconnect negative battery cable first to avoid spark. Failure to do so could result in death or serious injury. (W000238)

⚠ WARNING



Vision loss. Eye protection is required when servicing unit. Failure to do so could result in vision loss or serious injury. (W000377)

A 12 volt battery is required to operate the starter motor on Generac's 459 CC G26 G-Force Single engine. Battery rating should meet the minimum battery recommendations for best starter life and performance. See unit owner's manual for correct battery requirements.



020592

Figure 10-4. Typical Battery

Battery Cable Recommendations

Battery Cable Recommendations:

These cable sizes are based on total length of cable from positive (+) battery terminal to starter, plus ground return to negative (-) battery terminal.

- #4 AWG — 6 ft (1.8 m) or less
- #2 AWG — 12 ft (3.7 m) or less

Inspecting Battery

Proceed as follows to inspect battery:

1. Inspect and clean if necessary. Inspect for:
 - corrosion
 - dirt
 - terminals and clamps—good condition
 - terminals and clamps—secure connections
2. Fully charge battery.

IMPORTANT NOTE: DO NOT exceed charge rate of 1/10 ampere for every ampere of battery rating. Contact battery manufacturer for maximum charge recommendations. Use a trickle charger to reduce charge rate.

NOTE: Periodically unplug charger if battery becomes hot to the touch or emits excessive gas or electrolyte solution.

Battery Disposal

Always recycle batteries in accordance with local laws and regulations. Contact your local solid waste collection site or recycling facility to obtain information on local recycling processes. For more information on battery recycling, visit Call2Recycle website at: <http://Call2Recycle.org/locator>.

Section 11: Lubrication System

Description

The Generac 459 CC G26 G-Force Single engine uses a full pressure lubrication system with an oil filter. A gerotor oil pump draws oil from a screened oil pickup in the sump and pumps oil through the oil filter.

Filtered oil flows through an oil gallery in the sump and is distributed to the main bearings, connecting rod bearings, hydraulic lash adjusters, and camshaft bearings. Engine oil pressure varies with oil viscosity, ambient air temperature differences, operating temperatures, and engine load.

- Oil pressure - @ 70 °F (21 °C): 15 – 50 psi (103 – 345 kPa)

Protection Systems

Low Oil Pressure Switch

See [Figure 11-1](#). The engine is usually equipped with a normally open type, low oil pressure switch (A) which closes the circuit between the terminals when oil pressure drops below 5 +/- 2 psi (34 kPa). The engine will shut down on low oil pressure if one terminal is connected to the ignition ground wire and the other terminal is connected to ground. The low oil pressure switch can also be connected to an alarm or light to indicate low oil pressure, instead of shutting down the engine.

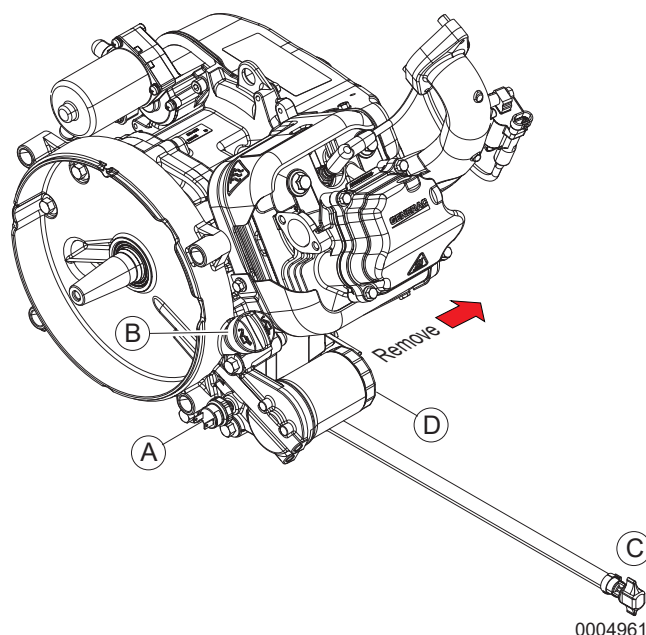


Figure 11-1. Low Oil Pressure Switch, Oil Fill, Oil Drain Plug, and Oil Filter

A delay built into the shutdown system on some engine applications allows oil pressure to build during starting. The delay allows the engine to run for about 10 seconds before sensing oil pressure. The engine shuts down if the engine is wired for low oil pressure shutdown and the system senses low oil pressure during operation. The engine will not run until 8 psi (55 kPa) of oil pressure is reached. If a restart is attempted within five seconds after shutdown, the engine may NOT start. The system needs 10 seconds to reset.

Checking Engine Oil Level



Engine damage. Verify proper type and quantity of engine oil prior to starting engine. Failure to do so could result in engine damage.

(C000135)

Oil capacity of the 459 CC G26 G-Force Single engine is approximately 1.4 qt (1.3 L). See [Specifications](#) for the recommended amount for each model.

Proceed as follows to check engine oil level:

1. See [Figure 11-1](#). Remove oil dipstick (B) and wipe it dry with a clean cloth.

2. See [Figure 11-2](#). Insert oil dipstick completely, then remove. Oil level should be at the FULL mark (A). If necessary, add oil until the FULL mark is reached. Do not fill above the FULL mark.

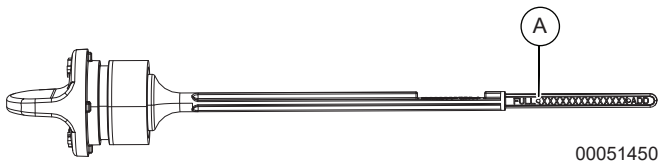


Figure 11-2. Oil Dipstick Full Mark

NOTE: Never operate engine with oil level below the ADD mark on oil dipstick. Doing so could damage the engine.

Changing Engine Oil And Filter

See [Specifications](#) for crankcase oil capacities. Do not use special additives. Verify unit is level when filling with oil. DO NOT OVERFILL.

The engine is equipped with an oil filter. Change oil and oil filter after the first eight hours of operation. Change oil and oil filter every 100 hours thereafter. In dirty or dusty operating conditions, or in extremely hot weather, change oil more often.

Proceed as follows to change oil while engine is still warm:

1. See [Figure 11-1](#). Clean area around oil drain plug (C).
2. Place suitable container below oil drain plug.
3. Remove oil drain plug and drain oil completely into a suitable container.
4. Install and tighten oil drain plug after oil is completely drained.
5. To change oil filter:
 - a. Locate oil filter (D).
 - b. Place a suitable container beneath oil filter and rotate filter counterclockwise to remove.
 - c. Coat gasket on new oil filter with clean engine oil. Install and rotate new oil filter clockwise until gasket contacts oil filter adapter, then tighten an additional three-quarters turn.
6. Remove oil fill cap and insert a clean funnel into oil fill opening. Fill crankcase with recommended oil until oil level is at FULL mark on oil dipstick.
7. Install oil fill cap after crankcase is filled to correct level. Start engine to fill oil filter. Verify and adjust oil level as needed.
8. Dispose of used oil in accordance with local, state, or national codes.

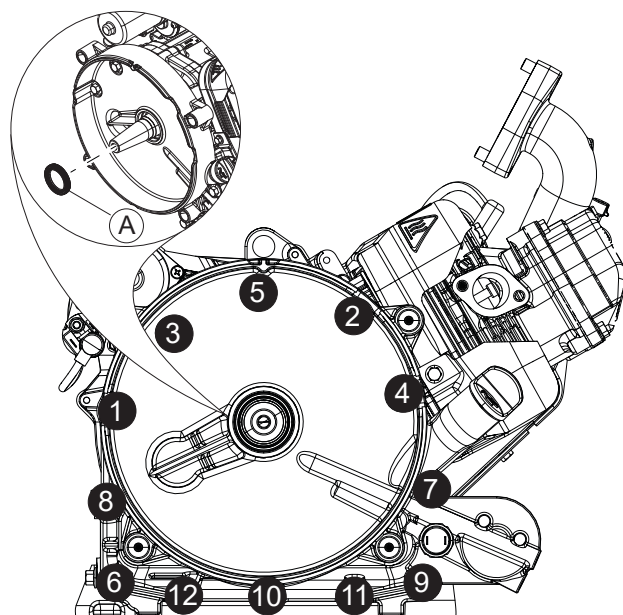
Section 12: Engine Assembly

Long Block Assembly

Installing Gear Cover

Proceed as follows to install oil seal and verify torque on gear cover assembly:

1. See [Figure 12-1](#). Press the crankshaft PTO oil seal (A) into the gear cover assembly (B) until oil seal is fully seated against bottom of bore.



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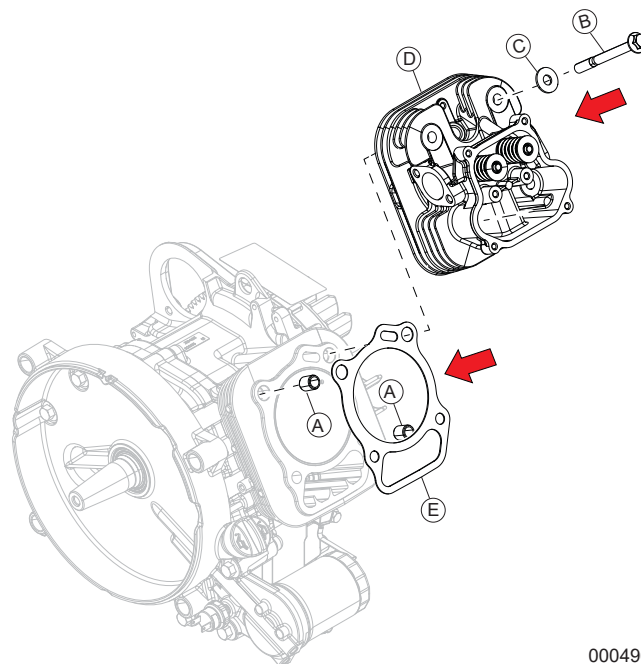
Figure 12-1. Install Oil Seal Into Gear Cover and Torque Sequence

2. Verify gear cover torque using the correct sequence shown. Final torque on gear cover is 32.4-39.6 ft-lb (43.9-53.7 Nm).

Installing Cylinder Head

Proceed as follows to install cylinder head:

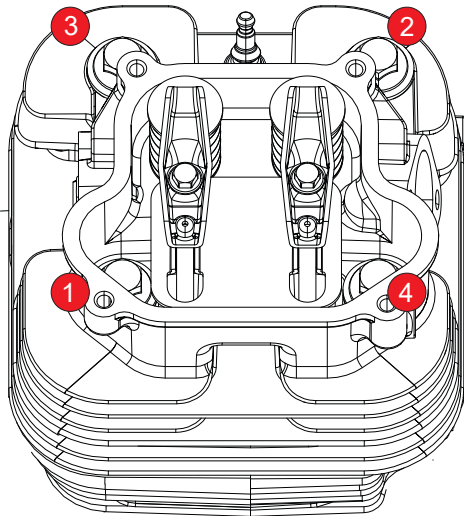
1. See [Figure 12-2](#). Install cylinder head bolt dowels (A) into cylinder head assembly.



00049618

Figure 12-2. Install Cylinder Head into Crankcase

2. Install cylinder head gasket (E) on engine long block. Install cylinder head (D) on long block assembly. Install four cylinder head bolts (B) and washers (C).
3. See [Figure 12-3](#). Tighten cylinder head bolts in pattern shown with the two step process as follows.
 - a. First sequence: tighten using pattern to 15–18 ft-lb (21–25 Nm).
 - b. Final sequence: tighten using pattern to 29.9–36.5 ft-lb (40.5–49.5 Nm).



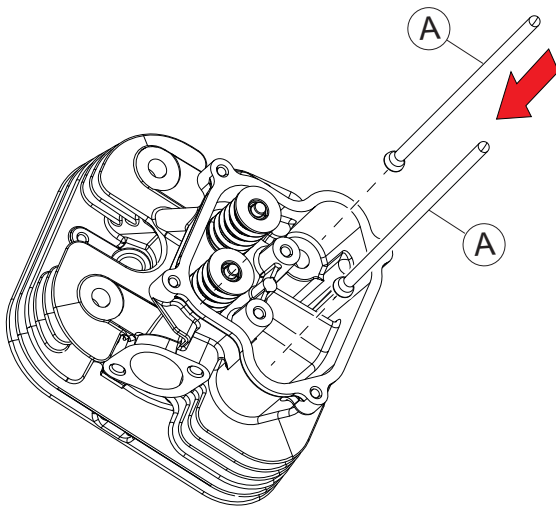
00051032

Figure 12-3. Cylinder Head Bolt Torque Sequence

Installing Valve Train

Proceed as follows to install valve train:

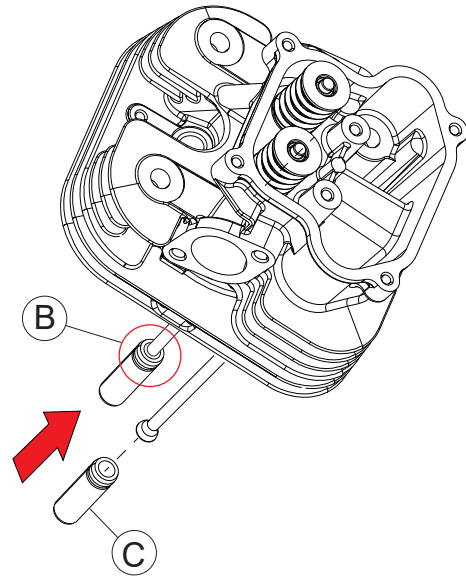
1. See [Figure 12-4](#). Apply assembly lube to each end of two pushrods (A). Install pushrods into cylinder one.



00051031

Figure 12-4. Install Pushrods into Cylinder Head

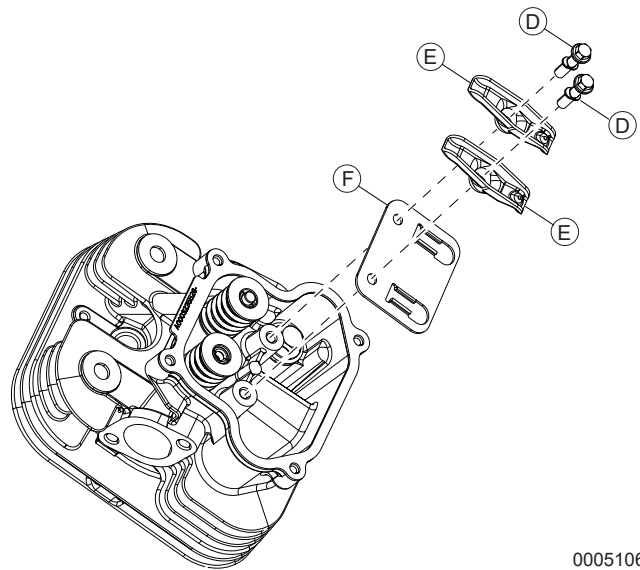
2. See [Figure 12-5](#). Verify each pushrod (B) is fully engaged in the recesses of the tappets (C).



00051030

Figure 12-5. Install Pushrods into Tappets

3. See [Figure 12-6](#). Install pushrod guide plate (F), rocker arms (E), and rocker arm pivot studs (D) loosely onto cylinder head. Verify ball end of pushrod tubes are inserted into ball recess in rocker arms.



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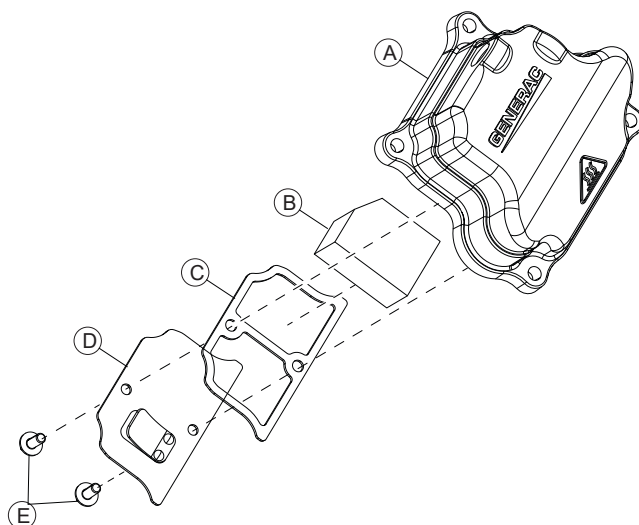
Figure 12-6. Install Rocker Arm into Cylinder Head

4. Rotate crankshaft clockwise, as viewed from flywheel side, until piston is at TDC on compression stroke (intake valve just closed).
5. Using fingers, spin pushrod while tightening pivot stud until resistance is felt on pushrod.
6. Tighten rocker arm pivot studs to 66.4-80.5 in-lb (7.5-9.1 Nm).
7. Repeat steps 3–5 for exhaust valve.

Installing Rocker Cover

Proceed as follows to install rocker cover:

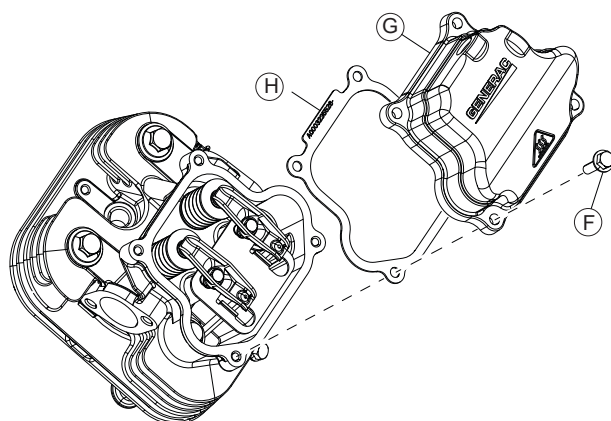
1. See [Figure 12-7](#). Assemble coalescing media (B), breather gasket (C), breather plate (D), and fasteners (E) into rocker cover (A) for cylinder to build breather assembly.



00051059

Figure 12-7. Install Oil Breather System into Rocker Cover

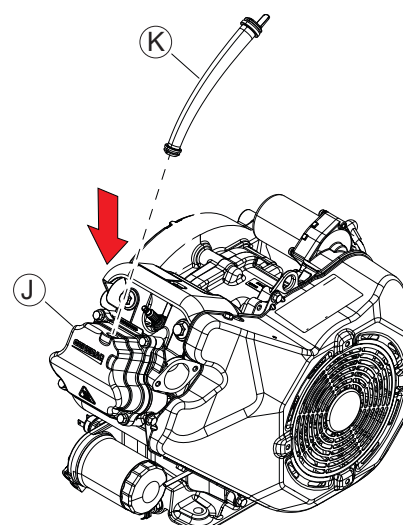
2. See [Figure 12-8](#). Install rocker cover gasket (H) onto cylinder. Use rocker cover bolts (F) to install rocker cover (G).



00051058

Figure 12-8. Install Rocker Cover Assembly onto Cylinder Head

3. Tighten rocker cover bolts to 57.5-66.4 in-lb (6.5-7.5 Nm).
4. See [Figure 12-9](#). Install breather hose (K) into rocker cover (J) facing opposite side of engine. Soapy water or spray silicone lubricant may be needed to fit breather hose into rocker cover hole.



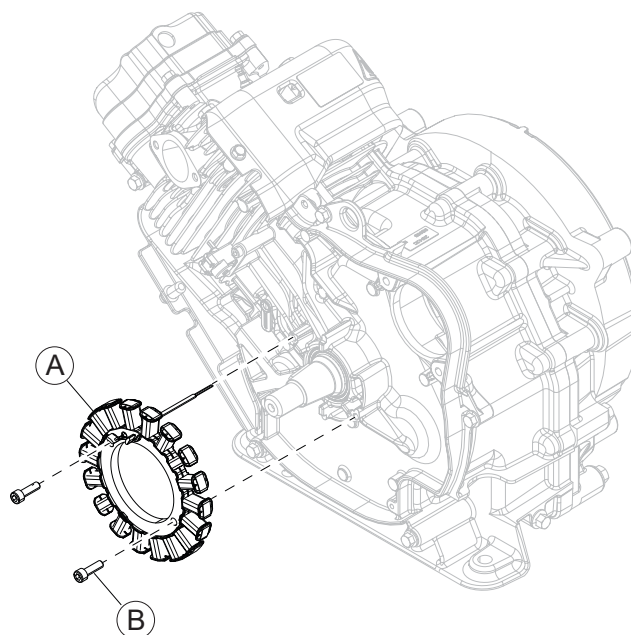
00051057

Figure 12-9. Install Breather Hose into Rocker Cover

Installing Ground Wire and Stator

Proceed as follows to install ground wire and stator:

1. Install ground wire connector of ignition coil onto backing plate.
2. See [Figure 12-10](#). Install clamp wire retainer (A) to crankcase with fasteners (B).



00049617

Figure 12-10. Install 1-phase Stator

3. Tighten fasteners to 53.1-57.5 in-lb (6-6.5 Nm).
4. Install 1-phase stator (C) using two fasteners (D).
5. Tighten stator fasteners to 97.4-120.4 in-lb (11-13.6 Nm).

Installing Cylinder Wrappers

Proceed as follows to install cylinder wrappers:

1. See [Figure 12-11](#). Install cylinder wrappers (A and B) to crankcase using four fasteners (C) (two fasteners per cylinder wrapper).

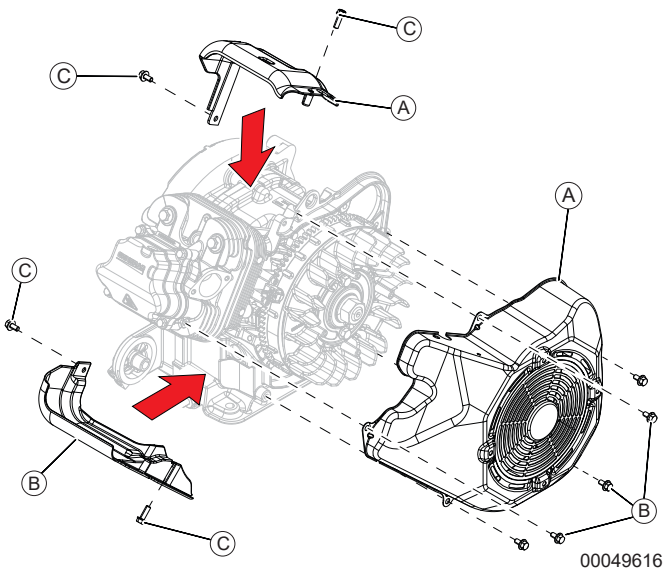


Figure 12-11. Install Cylinder Wrappers and Blower Housing

2. Tighten fasteners to 10 in-lb (1.13 Nm).

Installing Blower Housing

Proceed as follows to install blower housing:

1. See [Figure 12-11](#). Install blower housing (A) by fastening it to cylinder wrappers using five fasteners (C). Install j-clip (B) to area noted using appropriate blower housing fastener.
2. Tighten fasteners to 57.5-66.4 in-lb (6.5-7.5 Nm).

Installing Oil Filter and Oil Pressure Switch

Proceed as follows to install oil filter and oil pressure switch:

1. See [Figure 12-12](#). Apply small amount of clean engine oil to gasket on new oil filter (A).

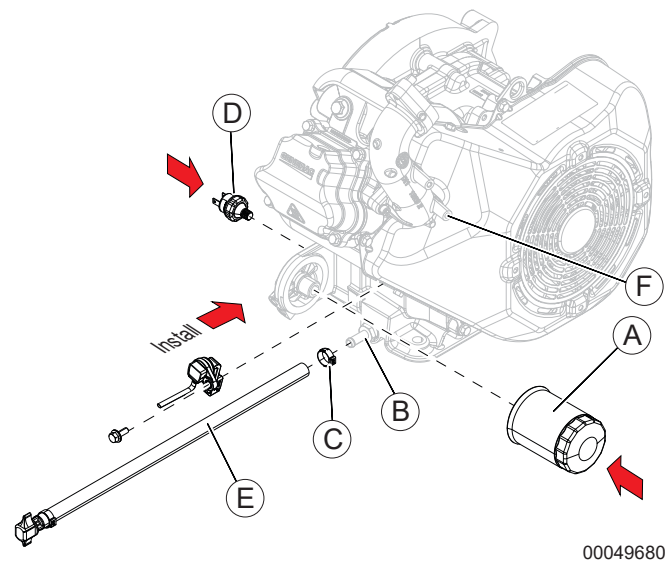


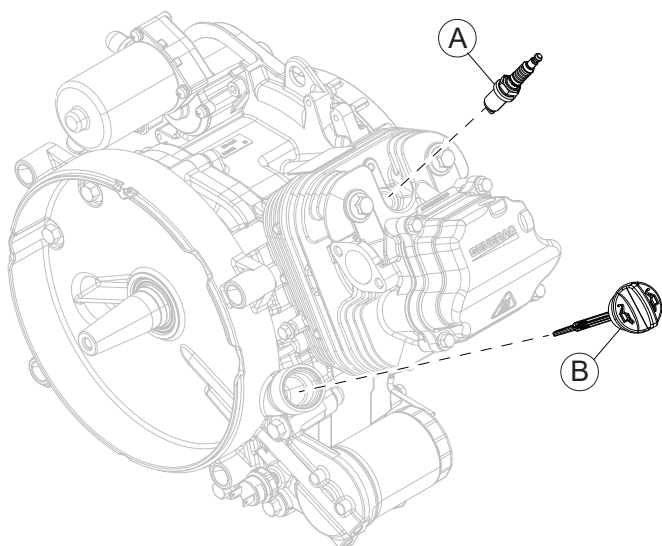
Figure 12-12. Install Oil Filter, Oil Pressure Switch, and Oil Drain Assembly

2. Screw onto threaded oil filter mount.
3. Tighten oil filter to 84 in-lb (9.5 Nm).
4. Verify oil pressure switch (D) has lock patch on threads.
5. Install switch in gear cover below oil filter.
6. Tighten switch two to three turns from finger tight.

Installing Oil Drain Line and Oil Dipstick

Proceed as follows to install the oil drain line and oil dipstick:

1. See [Figure 12-12](#). Verify 3/8 – 18 NPT oil drain barbed fitting (B) has lock patch on threads.
2. Install oil drain barbed fitting into crankcase.
NOTE: *NPT Fittings: As a general rule, pipe fittings with tapered threads should not be assembled to a specific torque. Instead, they should be installed by the number of turns past finger tight. All NPT plugs below 1 in size should be installed to 2-3 turns from finger tight.
3. Fit oil drain assembly (E) over barbed fitting and secure with 16.5 mm stepless ear clamp (C).
4. Hang oil drain hose in the j-clip on ignition coil mounting bolt (F).
5. Fill engine with 1.4 qt (1.3 L) of clean Generac 5W-20 synthetic engine oil (dry oil filter).
6. Apply small amount of clean engine oil to O-ring on dipstick.
7. See [Figure 12-13](#). Install oil dipstick assembly (B) into oil fill port.



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Figure 12-13. Install Dipstick and Spark Plugs

Installing Spark Plugs

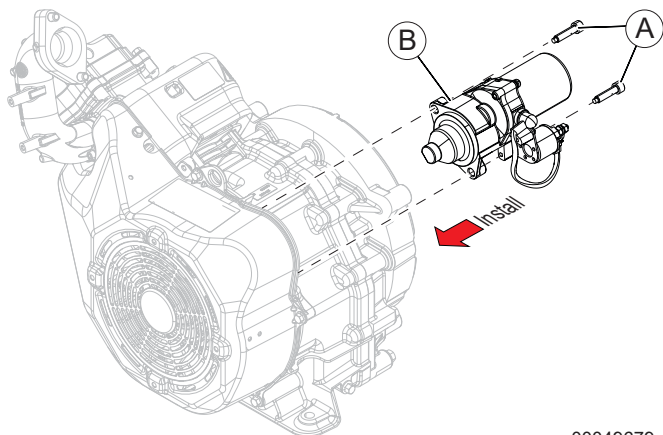
Proceed as follows to install spark plugs:

1. See [Figure 12-13](#). Install spark plugs (A) into both cylinders.
2. Tighten spark plugs to 14.8–22.1 ft-lb (20–30 Nm).

Installing Starter

Proceed as follows to install starter:

1. See [Figure 12-14](#). Install starter (A) onto crankcase with two fasteners (B) and two lock washers (C).



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Figure 12-14. Install Starter onto Crankcase

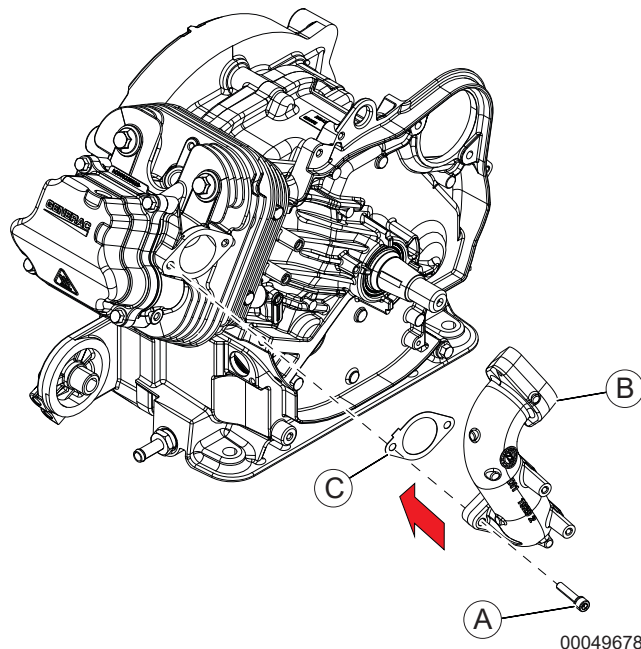
2. Tighten fasteners to 18.8–22.1 ft-lb (25.5–30 Nm).

Intake and Exhaust Assembly

Installing Intake Manifold

Proceed as follows to install intake manifold:

1. See [Figure 12-15](#). Place intake manifold gasket (C) onto cylinder head.



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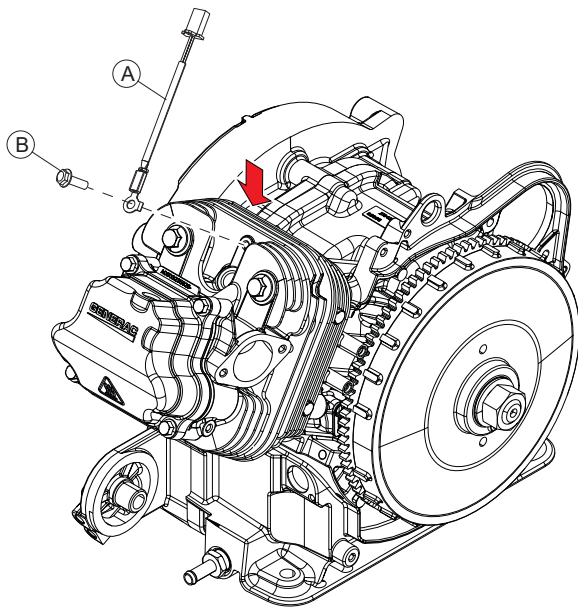
Figure 12-15. Install Intake Manifold onto Cylinder Head

2. Install intake manifold (B) onto cylinder heads using four fasteners (A).
3. Tighten fasteners to 88.5–101.8 in-lb (10–11.5 Nm).

Installing Head Temperature Sensor

Proceed as follows to install head temperature sensor:

1. See [Figure 12-16](#). Install head temperature sensor (A) with fastener (B).



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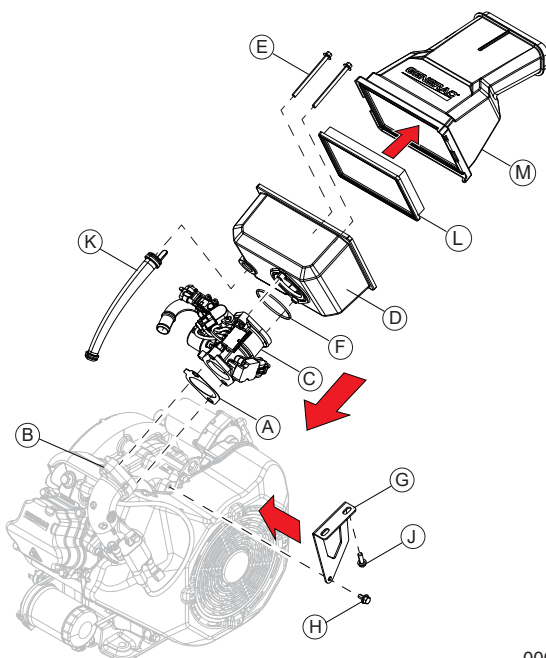
Figure 12-16. Install Head Temperature Sensor

2. Tighten temperature fastener to 57.5–66.3 in-lb (6.5–7.5 Nm).

Installing Throttle Body and Air Box

Proceed as follows to install throttle body and air box:

1. See [Figure 12-17](#). Install gasket (A) on intake manifold (B).



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Figure 12-17. Install Air Box, Box Belts, Breather, and Air Filter

2. Install throttle body assembly (C) and air box base (D) onto intake manifold using two bolts (E). Place gasket (F) between throttle body and air box.

3. Tighten bolts to 7.4–8.5 ft-lb (10–12 Nm).
4. Install air box support bracket (G) onto blower cover using mounting bolt (H).
5. Install air box support bolts (J) into bottom of air box.
6. Tighten air box support bolts to 10 in-lb (1.13 Nm).
7. Install breather hose (K) into mount in air box. Take care not to damage breather hose upon insertion.

Installing Air Filter

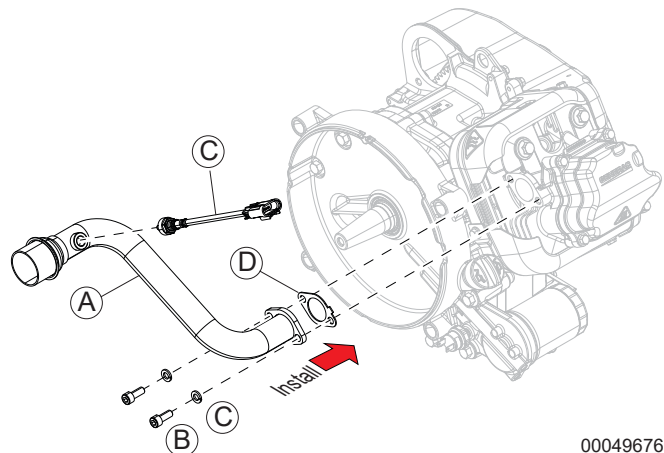
Proceed as follows to install air filter:

1. See [Figure 12-17](#). Install air filter (L) inside air box base (D).
2. Install air box cover (M) by hooking cover onto side of airbox base and fasten with snap latch on opposite side.

Installing Exhaust Manifold

Proceed as follows to install exhaust manifold:

1. See [Figure 12-18](#). Place exhaust manifold gasket (D) on cylinder head and install exhaust manifold (A) onto cylinder head using two washers (C) and fasteners (B).



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Figure 12-18. Install Exhaust Manifold

2. Tighten fasteners to 18 ft-lb (24.4 Nm).
3. Install O₂ sensor (C) in exhaust manifold.
4. Tighten O₂ sensor to 12.2–14.4 ft-lb (16.5–19.5 Nm).

Section 13: Troubleshooting

PROBLEM	POSSIBLE CAUSES	REMEDY
Engine does not start	Low oil level.	Replenish oil as necessary.
	Dirty air cleaner.	Clean or replace air cleaner element.
	Obstructed fuel line.	Clean, re-route, or replace fuel line.
	Fouled spark plug.	Replace spark plug. Inspect for cause of fouling (engine condition).
	Incorrect spark plug.	Replace with correct spark plug.
	Loose or faulty ignition wiring.	Inspect wiring; repair or replace.
	No ignition spark.	Inspect ignition system; replace faulty part(s).
	Incorrect valve clearance.	Inspect push rods, lifters, and rocker arms for damage (bent) or incorrect operation.
	Poor compression.	Inspect for worn or scored cylinder, burnt valve, burnt valve seat, or loose cylinder head bolts or spark plug.
Engine knocks	Carbon in combustion chamber.	Clean carbon from head or replace engine if excessive carbon also appears in the cylinder.
	Loose flywheel.	Inspect flywheel key and keyway; replace parts if necessary. Tighten flywheel nut.
	Worn cylinder.	Replace engine.
	Loose or worn connecting rod.	
	Incorrect valve clearance.	Inspect push rods, lifters, and rocker arms for damage (bent) or incorrect operation.
	Overloaded engine.	Reduce excessive load.
Engine misses under load	Incorrect spark plug.	Install correct spark plug.
	Fouled spark plug.	Replace spark plug. Inspect for cause of fouling (engine condition).
	Cracked porcelain on spark plug.	
	Incorrect spark plug gap.	Regap spark plug.
	Incorrect valve clearance.	Inspect push rods, lifters, and rocker arms for damage (bent) or incorrect operation.
	Weak valve springs.	Replace weak valve spring(s).
Engine lacks power	Incorrect spark plug.	Install correct spark plug.
	Incorrect valve clearance.	Inspect push rods, lifters, and rocker arms for damage (bent) or incorrect operation.
	Low oil level.	Replenish oil to correct level.
	Obstructed air cleaner.	Clean or replace air cleaner.
	Worn or burned valves or valve seats.	Grind valves and valve seats.
Engine overheats	Obstructed air flow.	Remove obstructions.

PROBLEM	POSSIBLE CAUSES	REMEDY
	Clogged cooling fins.	Clean cooling fins.
	Carbon buildup in combustion chamber.	Remove cylinder head and clean carbon.
	Overloaded engine.	Reduce excessive loading.
	Insufficient lubrication.	Replenish oil to correct level.
Engine vibrates excessively	Loose engine mounting bolts.	Tighten mounting bolts.
	Bent crankshaft.	Replace engine.
	Problem in equipment connected to engine.	Inspect equipment driven by engine.
Engine surges or runs unevenly	Intermittent ignition spark.	Test ignition as per Ignition section; repair or replace faulty parts as necessary.
Engine uses excessive amounts of oil	Oil level too high.	Drain excess oil from crankcase.
	Loose oil filler cap, or damaged oil filler cap gasket.	Tighten or replace fill cap as necessary.
	Damaged or dirty breather.	Clean or replace rocker cover and breather assembly as necessary.
	Clogged drain holes in breather.	Unclog drain holes in breather.
	Leaking gaskets or seals.	Replace gaskets or seals.
	Excessively worn valve guides.	Replace cylinder head.
	Worn cylinder walls.	Replace engine.
	Worn piston rings and/or grooves.	
	Obstructed oil passages.	Clean oil passages.
Crankshaft oil seal leaks	Dirty or faulty crankcase breather.	Clean or replace breather.
	Hardened or worn oil seal.	Replace engine.
	Worn crankshaft seal contact surface.	
	Worn crankshaft sleeve bearing.	
	Seal not seated squarely in bore.	
	New seal damaged during installation.	
	Bent crankshaft.	
Breather is passing oil	Oil level too high.	Drain excess oil from crankcase.
	Loose oil filler cap, or damaged oil filler cap gasket.	Replace cap gasket and tighten cap.
	Damaged or dirty breather.	Clean or replace breather as necessary.
	Clogged drain holes in breather.	Unclog drain holes in breather.
	Aligned piston ring end gaps.	Replace engine.
	Rings seated incorrectly.	

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